Title. A childhood gender gap in parent-to-child financial transfers

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Abstract. Parent-to-child financial transfers—which we label children's 'income'—are important for socialization and development[1,2], providing opportunities to learn about money, negotiate with parents, and express preferences through consumption. However, the nature of children's income has remained largely invisible because it typically consists of informal, unrecorded payments[3–6]. Here, we use 126M transfers to 1.15M UK children aged 6–15, linked to 101M child purchases, to study how children's income varies by gender and age and relates to consumption. We find a gender income gap in childhood: girls aged 6–10 receive 6–11% less money per month than boys. The gap then diminishes and reverses in adolescence, but model-based estimates imply girls do not reach cumulative income parity until age 15 and never achieve cumulative savings parity within the observed age range. The childhood gap primarily reflects girls receiving money less frequently than boys. In adolescence, this frequency difference narrows while transfer amounts increasingly favour girls. Much of children's income is consumption-related, and income changes across age track spending patterns that are consistent with gendered consumption norms. These findings reveal how financial gender inequality can emerge, accumulate and change across childhood and adolescence, well before labour-market entry.

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Introduction

Financial transfers from parents to children play an important role in socialization and development[1,2]. These transfers—which we label children's 'income'—provide opportunities to learn about money through spending and saving, negotiate with parents over specific purchases, and express nascent preferences and values through consumption choices. However, children's income has remained largely invisible because it typically takes the form of informal, unrecorded payments[3–6]. Thus, foundational questions about the nature of children's income remain unresolved.

Here, we use large-scale app data on parent-to-child financial transfers and child purchases to examine how children's income varies by gender, changes with age and relates to consumption. Gender differences emerge early in life[7,8] and are sometimes driven by parents[9–12]. Parents' choices can reflect beliefs about their children's preferences[13], and beliefs about gender differences in preferences are widely held[14]. Yet little is known about how financial gender inequality emerges, accumulates and changes across childhood and adolescence—or how gendered spending patterns might shape these outcomes.

We document a gender income gap in childhood (age 6–10) in which girls receive 6.36–10.97% less monthly income through the app than boys. The gap then narrows and reverses in adolescence (age 11–15). However, model-based estimates imply that girls do not reach cumulative income parity until age 15, and that they never achieve cumulative savings parity within the observed age range. The income results are present whether a child's account is administered by their mother or father and are similar for the 135,159 girl–boy sibling pairs in the data. These findings show that financial gender inequality can emerge well before labour-market entry, and thus well before most individuals make the career and family decisions that shape their

economic trajectory in adulthood[15–21]. They also help reconcile conflicting findings in survey-based studies of children's "pocket money" by showing how contemporaneous parity can conceal cumulative disadvantage[22–24].

The income gap in childhood is concentrated in ad hoc transfers for specific purchases and monetary gifts rather than regular allowances or task-specific payments, and it primarily reflects girls receiving money from their parents less often than boys. In adolescence, the frequency difference narrows while transfer amounts increasingly favour girls; girls also receive more allowance and task-related income. Allowances are paid in part to reward the completion of household chores[25,26], so the latter pair of observations is consistent with the possibility that the monthly income gap reverses in adolescence in part because girls perform more paid housework than boys. More broadly, this set of findings suggests that gendered income differences in our sample do not lie in consistently smaller payments, but in informal decisions by parents about when to provide money.

Spending records provide a complementary account of how the income gap changes from childhood to adolescence. Boys spend more on digital goods and services related to video games, which are frequent, relatively inexpensive purchases that are easy for even young children to make online. Girls spend more on clothes and accessories, which are less frequent, higher-value purchases that are often made in person, making them more feasible in adolescence than in childhood. These spending patterns track both the income gap in childhood and its reversal in adolescence because a substantial share of children's income is connected directly to their purchase requests or anticipated needs. Boys' relatively small but frequent purchases are consistent with them receiving more total income in childhood via smaller but more frequent income transfers. Girls' relatively large purchases are consistent with them receiving larger income transfers across

much of the age range we observe, and the relative ease of making these purchases in adolescence (versus childhood) is consistent with the income frequency gap narrowing and the total monthly income gap reversing. This correlation between income and spending also suggests a broader developmental implication: the distributional consequences of gendered consumption norms can change across life stages, contributing to inequality in one stage while helping reverse it in the next.

Data

The data come from a UK financial services firm whose subscription-based app lets parents transfer money to their children, who can then make purchases with an app-linked debit card. (For additional app details, see Methods, 'Data overview'.) We observe 1,154,275 children (50.51% female), 794,991 parent account holders (74.01% female), 126,357,734 parent-to-child payments and 100,865,877 child purchases. All transactions are recorded by the app rather than self-reported by users, similar to digital transactions used to study adults' income and spending[27–29].

Parents transfer income to children through one of four categories defined by the app. Ad hoc transfers (32,048,573) are generally tied to specific purchases children want or need to make. For example, parents will send an ad hoc transfer if they agree to a child's request for money to buy new clothes or a video game. Monetary gifts (18,921,071) are typically sent on birthdays and holidays, but they are also used to reward things like good grades at school or to celebrate events like graduation. Allowances (59,490,604) are recurring payments, often weekly, and the app suggests £0.50–£1.00 per week for each year of age. Task payments (15,897,486) reward children for completing specific chores or duties assigned by parents and can complement or replace allowances.

The data span June 2017 to February 2023. We aggregate transactions to a child-month panel, yielding 26,373,488 observations (Methods, 'Panel construction'). Mean (s.d.) monthly income is £40.60 (£86.67): £18.66 from ad hoc transfers, £11.75 from gifts, £9.56 from allowances and £0.63 from tasks. Mean (s.d.) monthly spending is £27.72 (£55.16). Mean (s.d.) monthly savings, calculated as app income minus spending (including cash withdrawals), is £12.88 (£60.10).

Extended Data Table 1 reports sample sizes and gender shares by age. Parents provided gender and age information when subscribing to the app. We treat children's age as a discrete variable ranging from 6 to 15 in our analysis, but for ease of exposition we refer to ages 6 through 10 as 'childhood' and ages 11 through 15 as 'adolescence'[30]. The provider anonymized the data before sharing them to ensure that no individual user is identifiable.

We proceed in three stages. First, we estimate age-specific gaps in monthly income and savings, and construct cumulative measures based on those estimates. Second, we measure the income gap in each category and decompose it into frequency and amount. Third, we examine whether gender- and age-specific spending patterns provide a coherent account of the income differences we observe.

Gender gaps in income and savings

Figure 1 and Extended Data Table 2a report the gender gaps in estimated monthly and cumulative income. Using regressions that control for parent characteristics, household child count, and year-month fixed effects (see Methods, 'Main regression'), we observe a gender income gap in childhood such that girls receive 6.36–10.97%, or £1.53–£3.11, less per month than boys (all $P <$

0.001). At age 10, for example, girls receive an estimated £27.23 per month compared with £30.33 for boys.

In adolescence, the monthly income gap narrows sharply at age 11 and reverses from age 12 onward. However, the estimates of cumulative income remain lower for girls than boys until age 15. The cumulative estimates sum annualized age-specific predicted monthly means and are therefore an implied age profile, not a trajectory observed for every child from 6 to 15. Nonetheless, they suggest it takes girls almost a full decade to reach cumulative income parity with boys during this formative life stage. Extended Data Table 2b shows that income differences exist whether a child's account is administered by their mother or father. Extended Data Table 2c shows similar results among the 135,159 girl–boy sibling pairs in the data, using household fixed effects to compare girls and boys in the same household.

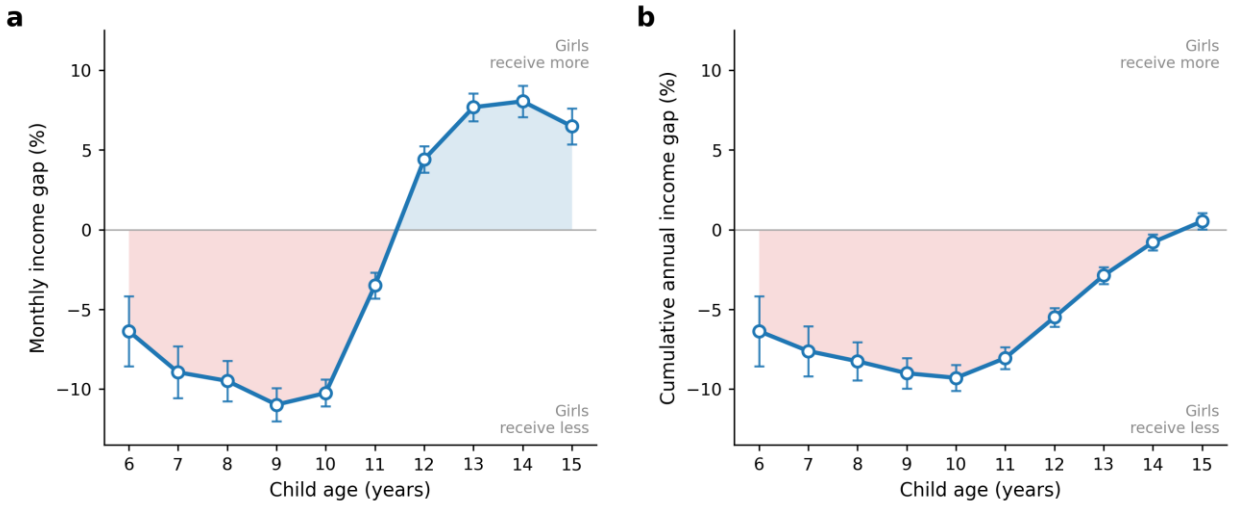


Fig. 1 | The gender income gap. a, Monthly income estimates. **b**, Cumulative income estimates. Y-axis values are percentage differences between girls and boys; negative values indicate that girls receive less. Absolute monthly differences range from −£3.11 to £4.86 and cumulative differences from −£157.51 to £24.74. Error bars show 95% confidence intervals. Full statistics are reported in Extended Data Table 2a.

One potential consequence of the gender income gap in childhood is that girls may save less than boys. To examine this possibility, we estimate monthly and cumulative savings using the same methods as for income. The results indicate that girls have significantly lower monthly savings through age 12 and lower cumulative savings at every age (Figure 2; Extended Data Table 3). At age 15, the cumulative savings gap remains £103.71, or 6.46% ($P < 0.001$), suggesting that girls never achieve cumulative savings parity within the observed age range. Extended Data Tables 4a and 4b show that girls and boys spend similar shares of income in childhood, which is consistent with the savings gap arising mainly from income differences. In adolescence, girls' spending share rises relative to boys', allowing the cumulative savings gap to persist even as the monthly income gap reverses.

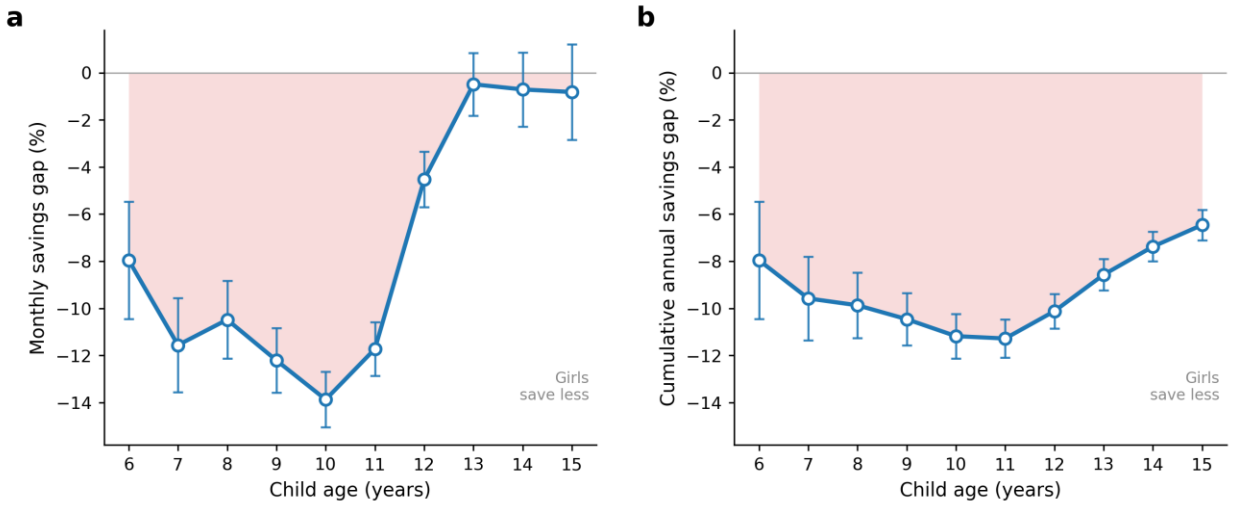


Fig. 2 | The gender savings gap. **a**, Monthly savings estimates. **b**, Cumulative savings estimates. Y-axis values are percentage differences between girls and boys; negative values indicate lower savings for girls. Absolute monthly differences range from −£1.63 to −£0.08 and cumulative differences from −£103.71 to −£11.69. Error bars show 95% confidence intervals. Full statistics are reported in Extended Data Table 3.

Income gaps by source

To better understand why there is a gender income gap in childhood—and why it reverses in adolescence—we next perform our main regression analysis for each income category. The results are presented in Figure 3 and Extended Data Table 5. Ad hoc transfer and gift income follow a pattern similar to that of total income. In childhood, girls receive 10.28–15.97% (£1.10–£2.08) less ad hoc income per month and 6.14–14.34% (£0.38–£1.05) less gift income per month (all $P < 0.004$). Both gaps then narrow and reverse in adolescence. Ad hoc transfers and monetary gifts account for approximately three-quarters of total income, so gender differences in total income can largely be attributed to differences in these categories. More generally, the size of these income streams shows that a large portion of total income is linked to children's anticipated or desired consumption. This is true for ad hoc transfers, which typically occur when parents transfer money in anticipation of a specific expense their child will encounter, or when a child asks for money to make a desired purchase. It is also likely true for at least some monetary gifts, which can reflect givers' beliefs about how recipients will spend the money[31,32].

Allowances, which comprise roughly one-quarter of total income, tell a different story: there are no significant gender differences in childhood, and girls receive 1.19% to 4.77% (£0.11 to £0.59) *more* monthly allowance income than boys in adolescence. One potential reason for this is that allowances are paid in part to reward the completion of household chores[25], and girls generally do more housework than boys[26,33]. The data do not allow us to directly confirm this possibility, but the app does note in its marketing to parents that one reason to use the app's allowance function is to reward children for housework. Moreover, we observe a similar pattern for task payments, which are made by parents when a child completes a specific chore or household task and are used as a complement to or substitute for allowances. Although task

payments generally constitute a small amount of total income, it is noteworthy that girls receive significantly more task payment income than boys from age 9 onward. Taken together, the results for allowance and task income suggest that girls may be reversing the income gap in adolescence in part by performing more housework than boys.

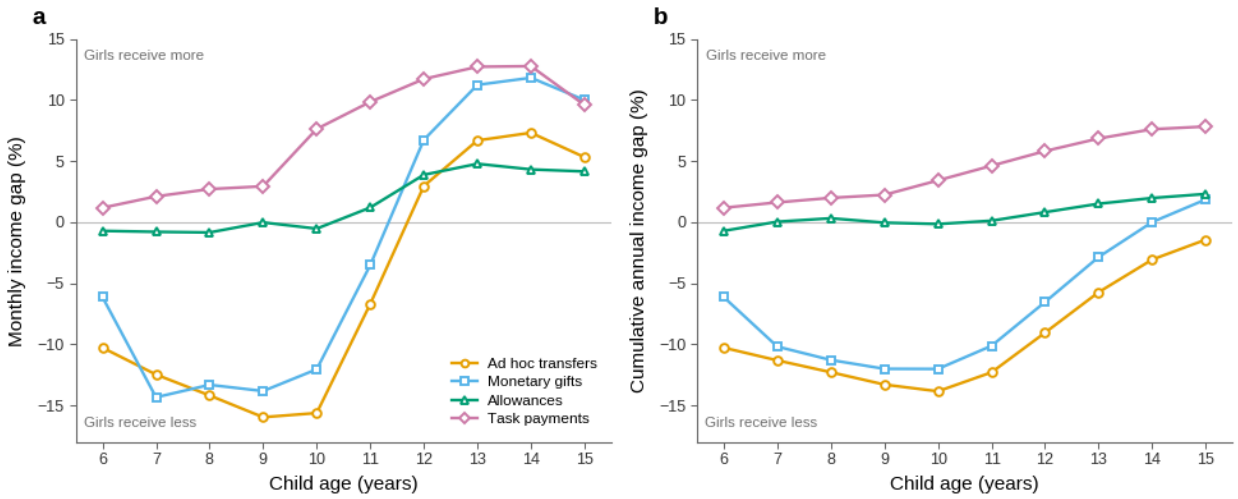


Fig. 3 | The gender income gap by category. a, Estimated monthly differences. **b**, Estimated cumulative differences. Y-axis values are percentage differences between girls and boys; negative values indicate lower income for girls. Lines show ad hoc transfers, gifts, allowances and tasks. Absolute monthly differences range from −£2.08 to £2.15, −£1.05 to £2.07, −£0.07 to £0.59 and £0.01 to £0.09, respectively. Absolute cumulative differences range from –£104.95 to –£13.22, –£55.02 to £24.49, –£0.68 to £25.24, and £0.08 to £5.80, respectively. Full statistics are reported in Extended Data Table 5.

Transfer frequency and amount

We next examine whether gender differences across income categories reflect how often parents transfer money or how much they transfer on each occasion. To do so, we perform our main regression analysis for the number of transfers per month and the average amount per transfer in each income stream. Results are presented in Figure 4 and Extended Data Table 6.

In childhood, girls receive 7.87% to 21.14% (0.06 to 0.20) fewer ad hoc transfers and 6.75% to 19.16% (0.03 to 0.09) fewer gift transfers per month (all $P < 0.001$), but these gaps narrow in adolescence. In contrast, the mean amount per ad hoc transfer is significantly higher for

girls than boys starting at age 8 (£14.56 versus £14.02, $P < 0.001$), the mean amount per gift transfer is significantly higher for girls starting at age 7 (£16.06 versus £15.55, $P = 0.002$), and both advantages grow with age. Because these categories dominate total income, the childhood income gap is primarily a frequency gap whereas the adolescent reversal involves converging frequency and larger average payments to girls.

Allowance frequency is broadly similar by gender. However, consistent with the preceding allowance analysis, adolescent girls receive 0.91% to 3.94% (£0.04 to £0.21) more per allowance payment than adolescent boys (all $P \leq 0.022$). Task payment frequency is higher for girls than boys starting at age 8, consistent with the observation that girls perform more housework than boys[26,33]. Differences in mean amount per task are small and not consistently significant, but tend to favour boys.

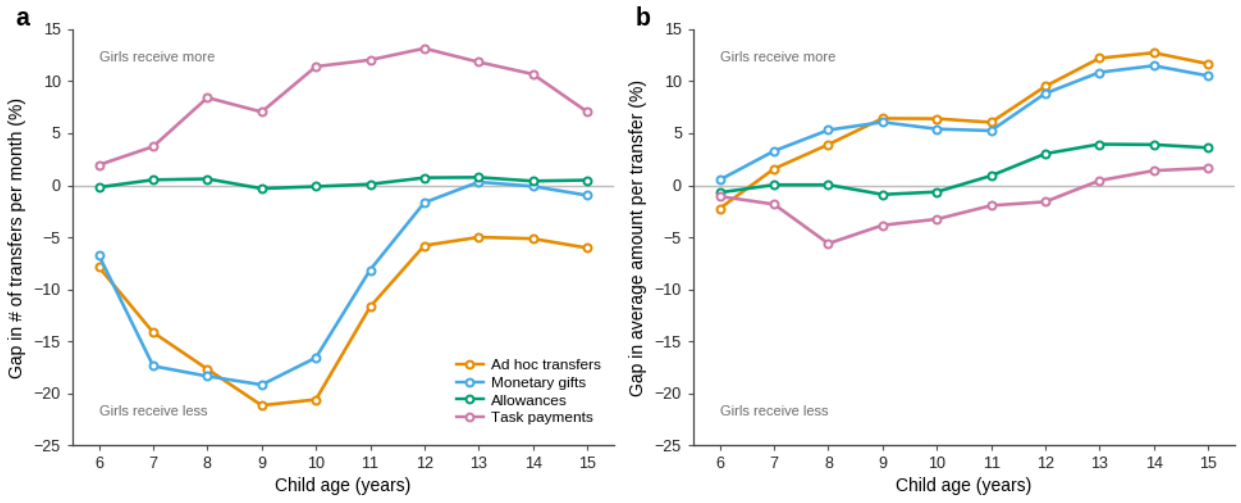


Fig. 4 | Gender gaps in income frequency and amount. **a**, Number of transfers per month. **b**, Mean amount per transfer. Y-axis values are percentage differences between girls and boys; negative values indicate lower values for girls. Lines show ad hoc transfers, gifts, allowances and tasks. Absolute differences in panel a range from –0.20 to –0.06 for ad hoc transfers, –0.09 to 0.00 for gifts, –0.01 to 0.02 for allowances, and 0.01 to 0.08 for tasks. Absolute differences in panel b range from –£0.30 to £2.06 for ad hoc transfers, £0.08 to £1.92 for gifts, –£0.03 to £0.21 for allowances, and –£0.05 to £0.02 for tasks. Full statistics are reported in Extended Data Table 6.

A consumption-linked account

The preceding analyses provide evidence that children's income is at least partly connected to their anticipated or desired consumption: the largest income stream is ad hoc transfers that are commonly used to fund specific purchases, and the second largest stream is monetary gifts that can be influenced by parents' beliefs about what children will purchase with their gift money. Thus, to further explain the gender differences in income we document above, we next examine children's consumption norms, which we define as their most common or typical spending behaviour. To do so, we first identify the merchants at which the children in our sample spend their money. We then use artificial intelligence to group spending into four broad categories: clothing and accessories, digital goods and services, food and beverage, and all other spending. For example, spending at merchants such as PlayStation and Xbox is categorized under 'digital goods and services', spending at H&M and Primark is categorized as 'clothing and accessories', and spending at McDonald's and Starbucks is categorized as 'food and beverage'. Finally, we calculate the amount each gender spends in each category at each age. (See Methods, 'Expense categorization procedure'.)

Figure 5 and Extended Data Table 7 present the four spending categories for boys and girls of each age, as represented by the mean amount they spend in each category per month. The presence of two gendered consumption norms is clear: boys consistently spend more than girls on digital goods and services (i.e., video game-related expenses), and girls consistently spend more than boys on clothing and accessories. This may help explain the gender income gap in childhood—and its reversal in adolescence—because these consumption norms differ in terms of spending frequency and amount as children age. On the one hand, playing video games is a daily habit for many children and adolescents[34,35], and in-game 'microtransactions' such as purchasing

extra lives are relatively cheap and easy to make online. On the other hand, shopping for clothing and accessories is more episodic, involves larger sums and often involves shopping in person, which makes it easier in adolescence than in childhood. Importantly, evidence of these differences is observable in the data: there are 2.90 digital purchases for every clothing purchase, the average amount of a clothing purchase is 2.42 times larger than that of a digital purchase (£17.53 versus £7.24, $P < 0.001$), and 99.14% of digital purchases are made online whereas 62.39% of clothing purchases are made in person ($P < 0.001$). Furthermore, monthly spending on digital goods and services is more strongly correlated with income transfer frequency in childhood than spending on clothing and accessories ($r_{clothing} = 0.10$, $r_{digital} = 0.26$), but this difference disappears in adolescence ($r_{clothing} = 0.16$, $r_{digital} = 0.16$). Additionally, the correlations between monthly spending in these categories and the mean amount of each income transfer are more similar to one another in childhood ($r_{clothing} = 0.17$, $r_{digital} = 0.11$) than in adolescence ($r_{clothing} = 0.27$, $r_{digital} = 0.14$). The contrast between digital and clothing spending therefore aligns with the distinction between income transfer frequency and amount: in childhood, digital spending is more closely related to income transfer frequency than clothing spending is, whereas in adolescence, clothing spending is more closely related to income transfer amount than digital spending is. These patterns may help explain why the income gap changes with age and the different combinations of income transfer frequency and amount that underlie it.

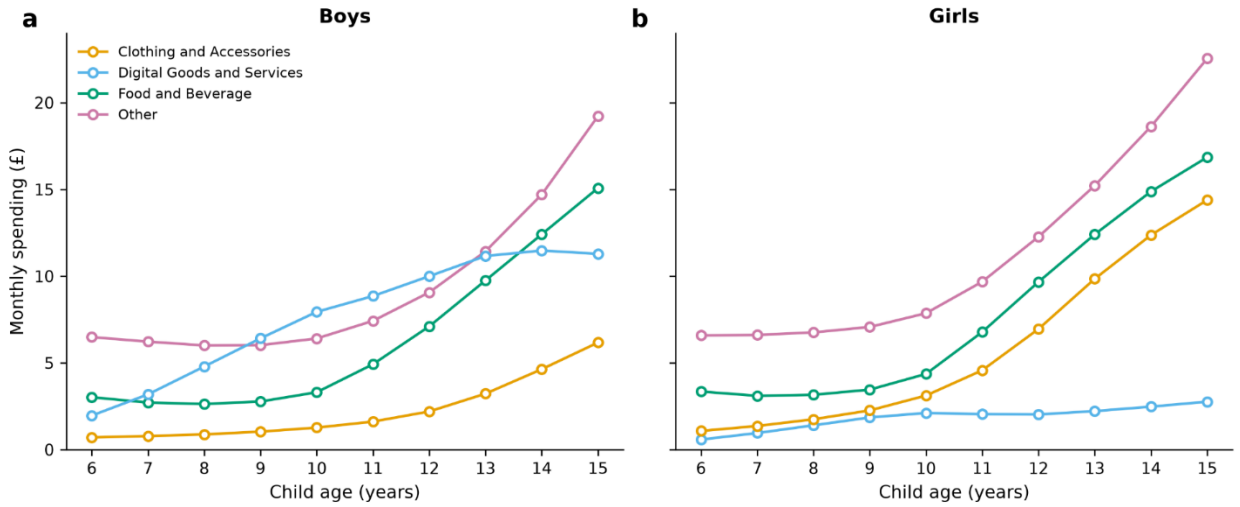


Fig. 5 | Gendered spending patterns. Mean monthly spending by age for clothing and accessories, digital goods and services, food and beverage, and all other categories. **a**, Boys. **b**, Girls. Full statistics are reported in Extended Data Table 7.

Income and spending during lockdown

The UK's initial COVID-19 lockdown sharply restricted in-person shopping but not online spending[36]. We use this disruption to ask whether income and spending moved in the direction predicted by the consumption-linked account: if gendered consumption norms help shape children's income, the lockdown should decrease girls' income more than boys' because children could no longer shop for clothes in person, but they could still play online video games.

We use regression analysis to estimate changes in average weekly income and spending during the lockdown (23 March–1 June 2020) as compared to the previous fifteen months (1 January 2019–22 March 2020) (see Methods, 'COVID-19 analysis'). Figure 6 and Extended Data Table 8 show that girls' average weekly income fell by approximately six times more than boys' during the lockdown (–£0.47 versus –£0.08 per week, $P < 0.001$), and this is almost entirely attributable to changes in ad hoc transfer income (–£0.55 versus –£0.16 per week, $P < 0.001$), which is the income stream most directly related to consumption. Girls' total spending also fell

more than boys' (–£1.90 versus –£1.13 per week, $P < 0.001$), because girls' larger decrease in in-person spending (–£2.42 versus –£1.41 per week, $P < 0.001$) was not fully offset by their increase in online spending (£0.96 versus £0.68 per week, $P < 0.001$). Taken together, these results are consistent with the possibility that gendered consumption norms shape differences in children's income: when the lockdown disrupted forms of consumption that are more common among girls than boys, girls experienced larger declines in both spending and ad hoc transfer income.

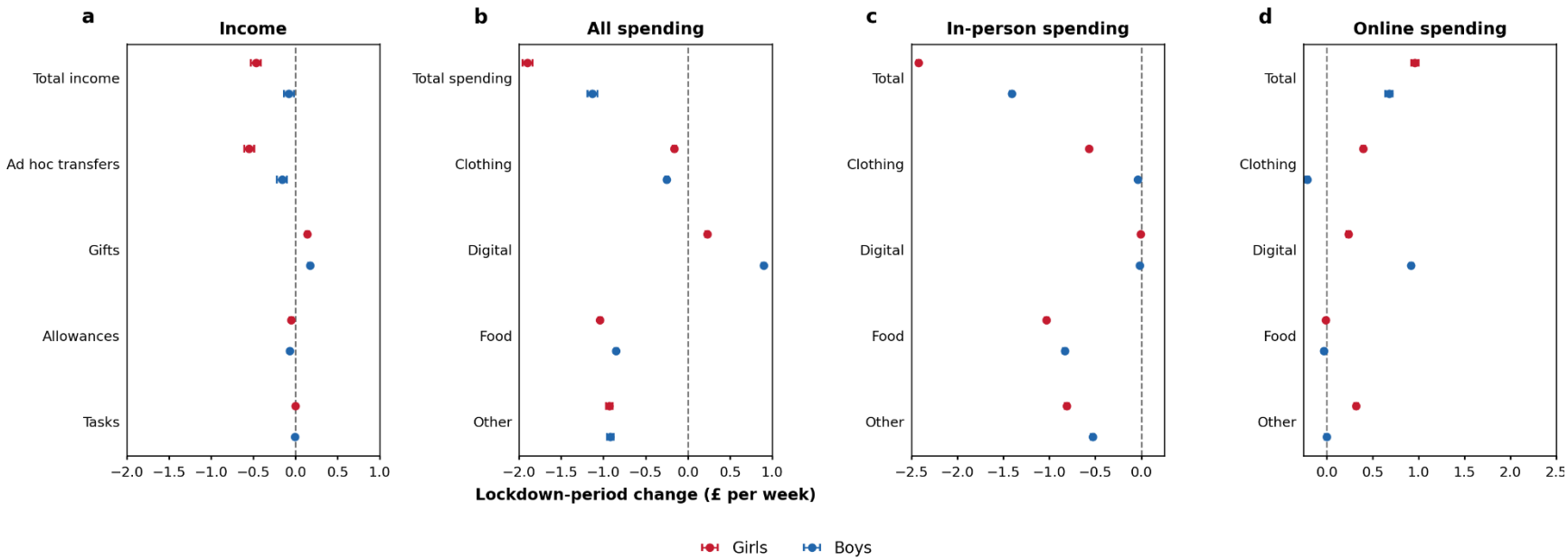


Fig. 6 | Lockdown-period changes in income and spending. Coefficients compare girls' and boys' weekly outcomes during 23 March–1 June 2020 with 1 January 2019–22 March 2020; error bars show 95% confidence intervals. **a**, Total and category-based income. **b**, Total and category-based spending. **c**, Total and category-based in-person spending. **d**, Total and category-based online spending. Full regression results are reported in Extended Data Table 8.

One limitation of this analysis is that the lockdown disrupted more than children's consumption opportunities[37]. Other lockdown-related changes may therefore have contributed to the observed patterns. A second limitation of this analysis is that it does not directly link income and spending. To address this, we next return to the age-specific income gaps and examine how they relate to gender differences in spending across all four categories.

Spending and income decomposition

To further examine the relationship between spending and income, we estimate descriptive mediation models for each age from 6 to 15 (Extended Data Figure 1; Methods, 'Mediation analysis'). The models separate the gender difference in monthly income into components associated with gender differences in spending in each of the four expense categories, and the remaining difference after controlling for spending. Following convention, we label these components 'indirect' and 'direct' effects, respectively.

The direct effect of gender on total monthly income is negative at every age and statistically significant through age 14 (Extended Data Table 9a). Thus, the income gap reversal in adolescence is no longer present after controlling for spending in all four categories. This is consistent with the possibility that consumption norms play a role in determining children's income.

The indirect effect of digital spending is negative and statistically significant at every age (Extended Data Table 9a), reflecting boys' higher spending on digital goods and the positive association between digital spending and income (Extended Data Table 9b). The indirect effects of clothing, food and beverage, and 'other' spending are positive and statistically significant from age 6 or 7 onward, reflecting girls' higher spending in these categories and their positive associations with income. Digital spending has the largest indirect effect in absolute terms at every

age. Clothing has the largest positive indirect effect from age 10 onward, with estimates increasing sharply in adolescence, from £2.73 at age 11 to £7.40 at age 15. This is consistent with the prominence of these two gendered consumption norms and suggests that girls' higher clothing spending may help explain why the income gap reverses in adolescence.

Taken together, the models show that gender differences in spending are associated with boys' higher monthly income in childhood and girls' higher monthly income from age 12 onward. At these older ages, the income differences associated with girls' higher spending on clothing, food and beverage, and 'other' items collectively exceed the opposing difference associated with boys' higher digital spending. At age 12, for example, girls receive an estimated £1.87 *more* per month than boys before controlling for spending, but £0.94 *less* when spending in all four categories is held constant. The £2.81 difference between these estimates is the net income difference associated with their spending patterns. These analyses are correlational rather than causal, but the patterns are consistent with the possibility that the same gendered consumption norms play a role in both the childhood income gap and its reversal in adolescence.

Discussion

We provide large-scale evidence that financial gender inequality exists in childhood. Girls receive less income than boys from parents as early as age 6, and although the monthly income gap reverses in adolescence, our model-based estimates suggest (a) girls do not reach cumulative income parity until age 15, and (b) girls never reach cumulative savings parity within the observed age range. These findings extend research on financial gender inequality backward in the life cycle, showing that disparities can arise well before labour-market entry. Furthermore, these differences arise during a period when children's preferences for fair resource allocation are developing[38,39].

Our results also suggest a different way to understand children's income. Whereas standard economic models typically treat consumption as following from anticipated income[40,41], our findings indicate that children's finances can operate in the reverse direction, with income following from anticipated consumption. A large share of children's income comes through ad hoc transfers tied to specific purchases, meaning that what children want or are expected to buy can shape how much and how often parents transfer them money.

Our consumption-linked account of children's income has the potential to help explain both the emergence and reversal of the gender income gap. Boys' relatively high spending on digital goods is frequent, inexpensive, and easy to conduct online at any age, whereas girls' relatively high spending on clothing and accessories is more episodic, higher value, and easier to conduct in person during adolescence than childhood. These gendered consumption norms therefore differ not only in cost, but also in frequency and timing. As a result, they are associated with different transfer frequencies and amounts across ages, consistent with the possibility that they contribute to both the monthly income gap in childhood and its reversal in adolescence.

Several patterns converge on this interpretation. Children's income is concentrated in ad hoc transfers and monetary gifts. The childhood gap is associated primarily with transfer frequency, whereas the adolescent reversal reflects a narrowing frequency gap combined with larger average transfers to girls. The COVID-19 analysis provides evidence that the lockdown disrupted both girls' spending opportunities and their income through the stream most closely tied to purchase requests, and the mediation analysis shows that spending categories account for part of the gender-income relationship. The results for allowance and task income also suggest that girls may be reversing the income gap in adolescence in part by performing more housework than boys.

These findings have practical implications for children's financial socialization. Prior work has emphasized allowances as a tool for teaching children about money[2,31,42], but our results suggest that parent-child negotiations around consumption may play a larger role in children's everyday financial experience. More regular and transparent transfers may reduce the extent to which children's income depends on gendered consumption norms, although such interventions should be tested directly.

Our analysis is not without limitations. For example, we cannot observe parents' non-monetary investments in children, such as lessons or durable goods. Thus, it could be the case that although girls receive less income than boys in childhood, they may receive more resources from their parents in other areas. If true, however, it is not clear that an advantage in one resource justifies a disadvantage in another, nor does it change the fact that girls end up with fewer opportunities to make their own financial decisions in childhood, which may help explain gender differences in financial confidence later in life[43].

We also cannot observe whether parents make cash transfers to children outside the app. Prior studies suggest that if such transfers do occur, they may disproportionately favour boys[44,45], which would make the income gap in childhood larger than we observe, and the reversal in adolescence smaller. However, there are strong reasons to believe that if such transfers occur in our sample, they are relatively small and infrequent. For example, the app has several advantages over using cash that help explain its popularity, including the ability to monitor children's spending, in-built financial education for children, and the assurance that children's money cannot be lost or stolen. Additionally, parents pay a subscription fee for the app, which gives them an economic incentive to use it. Nonetheless, future research could directly compare the nature of app-based versus cash transfers from parents to children.

Our findings are also limited to app users in the UK. However, we hypothesize that the income gap we observe in childhood likely underestimates differences in the broader population for two reasons. First, the parents in our sample pay for an app designed to help their children learn about money regardless of child gender. This suggests they may be relatively progressive in their approach to financial socialization, and thus less likely to discriminate. Second, the UK ranks fourth in the world in overall gender parity[46], which suggests the childhood gender gap in our sample may be smaller than in other countries. Nonetheless, the generalizability of our results across countries is an empirical question, and future research should examine whether similar gaps exist in different populations.

More broadly, our findings show that gender inequality can emerge through ordinary household financial interactions long before children enter the labour market. They also suggest the same consumption patterns can advantage boys at one stage and girls at another. However, reversal is not repair: later income parity does not restore girls' lost opportunities to exercise financial discretion at earlier ages, nor does it appear to close the savings gap. Thus, understanding the early roots of financial gender inequality requires tracking not only how much children receive, but when, why and how often.

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Methods

Data overview

Our analysis uses transaction-level data from a UK financial services firm. Its subscription app and linked debit card let parents transfer money, assign paid tasks and monitor activity; children manage balances and make purchases. Account creation requires parental verification and age-based merchant controls. Monthly subscription fees ranged from £2.49 to £3.99 per child during 2017–2023. The provider removed personally identifiable information before sharing the data and supplied each child's age and gender category, as entered by parents when signing up for the app.

Parents can send money to their children through four categories: (1) ad hoc transfers, generally linked to a specific need or purchase request; (2) gifts for occasions or achievements; (3) recurring allowances, usually weekly; and (4) task-specific payments, which parents assign and release upon completion of household chores or responsibilities. The app assigns these labels

consistently across users. The linked debit card records merchant identifiers, amounts and time stamps on established payment networks, allowing spending to be grouped by merchant category. The Durham University ethics review board approved this study (DUBS-2023-02-04T20:16:26-hvwk37).

Panel construction

The dataset covers all app activity from June 2017 to February 2023: 126,357,734 parent-to-child payments (32,048,573 ad hoc transfers, 18,921,071 gifts, 59,490,604 allowances and 15,897,486 task payments) and 100,865,877 child purchases. Parent-to-child income transfers include date, amount, child identifier and income category; child purchases include merchant identifier, amount and timestamp. Children are linked to the parent account for their household and are observed from ages 6 to 15. We treat age as discrete in our analysis but refer to 6–10 as childhood and 11–15 as adolescence for ease of exposition in our writing[30].

We aggregate transactions to the child-month level, including zero-payment months, to form 26,373,488 observations in which each child appears continuously from first to last recorded app activity. Results remain consistent when zero-payment months are excluded. For each child-month we calculate total income and spending through the app, as well as the amount received through each income category and the amount spent in the spending categories described below. Savings equals app income minus spending, including debit card purchases and cash withdrawals. Thus, it captures the portion of app income not spent or withdrawn through the app. Results are also consistent when cash withdrawals are excluded from spending.

Main regression

To measure gender differences in children's income, we construct a child-month panel dataset and estimate the following regression model:

$$y_{h,i,t} = \alpha + \sum_{a=7}^{15} \theta_a I(\text{ChildAge}_{i,t} = a) + \sum_{a=6}^{15} \gamma_a I(\text{ChildAge}_{i,t} = a)\text{Female}_i + \beta_1 \text{ParentAge}_{h,t} + \beta_2 \text{ParentAge}_{h,t}^2$$
$$+\beta_3 \text{ParentFemale}_{h,t} + \sum_{k\geq 2} \beta_{4,k} I(\text{ChildrenInHH}_{h,t} = k) + \delta_t + \varepsilon_{h,i,t}$$

Here, $y_{h,i,t}$ is total monthly income, measured in pounds sterling, received by child i in household h during month t. Total monthly income is defined as the sum of ad hoc transfers, monetary gifts, allowances, and task-specific payments. $I(\cdot)$ is an indicator function equal to one when the condition in parentheses is true and zero otherwise. $Female_i$ is an indicator equal to one for girls and zero for boys. $ChildAge_{i,t}$ is entered as a set of age indicators, allowing income to vary flexibly across ages 6 through 15.

The first summation estimates the age profile of income for boys, with age 6 omitted as the reference age. The second summation interacts each child-age indicator with child gender. The coefficients of primary interest are γ_a, which estimate the conditional difference in monthly income between girls and boys at age a, for ages 6 through 15. Thus, γ_a can be interpreted as the gender income gap at age a, conditional on parent characteristics, household composition, and year-month fixed effects.

The parent-level variables are measured at the household-month level. $ParentAge_{h,t}$ is the age of the parent associated with household h in month t, and $ParentFemale_{h,t}$ is an indicator

equal to one if that parent is female and zero otherwise. We include parent age and its square to allow for a nonlinear relationship between parent age and children's income. We include parent gender to control for potential differences in children's income associated with whether the household's parent account is held by a female or male parent.

$ChildrenInHH_{h,t}$ is the number of children in household h with an app account in month t. We enter this variable as a set of indicator variables, where k indexes the number of child accounts in the household-month. Single-child households with $k = 1$ are omitted as the reference category. Although many children in the data are the only child with an app account in their household, some households contain multiple children with accounts. We therefore index observations by child, household, and month, and control for the number of child accounts in the household-month.

The year-month fixed effects, δ_t, absorb seasonality, macroeconomic conditions, platform-wide changes, and other common shocks affecting all children in a given month. Standard errors are clustered at the household level to allow for arbitrary correlation in residuals among children in the same household and across repeated monthly observations from the same household.

Using the fitted model, we estimate predicted monthly income separately for girls and boys at each age from 6 to 15, holding the remaining covariates at their sample means within each age group. We then calculate the gender gap at each age in both absolute terms, measured in pounds sterling, and percentage terms, using boys' income as the denominator. For cumulative annual income, we multiply each age-specific predicted monthly value by 12 and sum over the relevant age range. We repeat the same procedure with savings as the dependent variable to obtain monthly and cumulative savings profiles. In both cases, cumulative estimates should be interpreted as

model-based summaries of the predicted age profile rather than as direct within-child longitudinal estimates.

The standard errors for the estimated gender differences are calculated using the delta method, which accounts for the covariance structure of the regression coefficients. Statistical significance is assessed using two-sided tests.

Expense categorization procedure

We map individual purchases to broad expense categories using Claude Opus 4.1, followed by manual author review. Each spending transaction contains a merchant identifier provided by the payment network that describes the type of merchant (e.g., 'Family Clothing Stores' or 'Grocery Stores, Supermarkets'). We observe 77,628 unique merchants, which are grouped into 315 merchant categories by the data provider. We use Claude to map each unique merchant identifier to a mutually exclusive spending category. Examples include Primark, ASOS, H&M and Zara as clothing and accessories; Xbox, PlayStation, Steam, Roblox and Fortnite as digital goods and services; McDonald's, Subway, Greggs, and Nando's as dining out; and Tesco, Sainsbury's, and Aldi as groceries and food. Merchants that could not be clearly assigned to a specific category were classified as Miscellaneous. The author team then manually reviewed all the category assignments and made one manual adjustment: "Petroleum and Petroleum Products" was reclassified as Groceries and Food since most transactions in this category were for small petrol-station purchases, which, for children and adolescents, more likely represent snacks and drinks than fuel purchases.

There were 13 categories in total: (1) Clothing and Accessories, (2) Digital Goods and Services, (3) Dining Out, (4) Groceries and Food, (5) Books and Stationery, (6) Education, (7) Entertainment and Recreation, (8) Gifts and Specialty Items, (9) Health and Beauty, (10) Home

and Garden, (11) Travel, (12) Cash, and (13) Miscellaneous. We retained Clothing and Accessories and Digital Goods and Services in their original form because they are clearly distinct categories. We combined Dining Out and Groceries and Food into a single Food and Beverage (F&B) category because they serve the same physiological needs. Spending in each of the remaining nine categories was comparatively small or, in the case of cash withdrawals, could not be assigned to a specific spending category, so we consolidated them into a single 'other' category.

For each child-month, we calculated total monthly spending in each category (Clothing and Accessories, Digital Goods and Services, Food and Beverage, and Other) by summing the transaction amounts for all transactions assigned to that category, including cash withdrawals in 'other'. Figure 5 plots estimated mean monthly spending in each category by child gender and age. Extended Data Table 7 presents the means and the corresponding gender gaps.

COVID-19 analysis

We examine weekly changes around the first UK COVID-19 lockdown, beginning 23 March 2020. The analysis uses 22,647,924 child-week observations from 1 January 2019 to 1 June 2020 and compares income and spending before and during the lockdown, when stay-at-home measures disrupted some consumption channels more than others. Because household finances and many other constraints changed simultaneously, coefficients should be interpreted as descriptive lockdown-period changes, not isolated causal effects of consumption restrictions. We suppress the household index h below for readability.

COVID-19 model specification. We estimate the following fixed effects regression:

$$y_{i,t} = \alpha_i + \delta_t + \beta_1 COVID_t + \beta_2 (COVID_t \times Female_i) + \varepsilon_{i,t}$$

where $y_{i,t}$ is the outcome (income or spending) for child i in week t, α_i represents child fixed effects, δ_t represents year-month fixed effects, and $COVID_t$ is an indicator equal to one for observations on or after 23 March 2020. The coefficient β_1 represents the estimated lockdown-associated change for boys, while $\beta_1 + \beta_2$ captures the corresponding change for girls. The coefficient of primary interest is β_2, which represents the difference between these changes. Because year-month fixed effects absorb common changes between months, the common lockdown-level component relies on within-month variation in lockdown status. The gender difference additionally uses changes in the girl–boy gap across the pre-lockdown and lockdown periods.

Standard errors are clustered at the household level. The inclusion of child fixed effects controls for all time-invariant characteristics of individual children. Year-month fixed effects absorb common time trends affecting all children, including seasonality, macroeconomic conditions, and platform-wide changes in user behaviour.

Outcomes. For income, we examine estimated changes in five categories: total income, ad hoc transfers, monetary gifts, allowances, and task payments. For spending, we examine estimated changes by channel (all, in-person only, and online only) and category (total, clothing and accessories, digital goods and services, food and beverage, and other).

Mediation analysis

We use structural-equation modelling (SEM) to estimate descriptive mediation models for each age from 6 to 15 (Extended Data Figure 1). Each model includes child gender as the independent

variable, total monthly income as the dependent variable, and monthly spending in the four expense categories as parallel mediators. The models separate the gender difference in monthly income into components associated with gender differences in spending and the remaining difference after controlling for spending. Following convention, we label these components 'indirect' and 'direct' effects, respectively. We estimate separate models to allow these relationships to vary with age, and we interpret these effects as descriptive associations rather than causal effects.

Mediation model specification. Each model consists of five linear equations. We use i to index children and t to index months, suppressing the household index for readability. The first four equations model spending in each category as a function of child gender and the control variables:

$$M_{1,i,t} = \alpha_1 + a_1 Female_i + X_{i,t}\lambda_1 + \varepsilon_{1,i,t}$$
$$M_{2,i,t} = \alpha_2 + a_2 Female_i + X_{i,t}\lambda_2 + \varepsilon_{2,i,t}$$
$$M_{3,i,t} = \alpha_3 + a_3 Female_i + X_{i,t}\lambda_3 + \varepsilon_{3,i,t}$$
$$M_{4,i,t} = \alpha_4 + a_4 Female_i + X_{i,t}\lambda_4 + \varepsilon_{4,i,t}$$

Here, $Female_i$ equals 1 for girls and 0 for boys. M_1 through M_4 represent monthly spending, in pounds sterling, on clothing and accessories, digital goods and services, food and beverage, and 'other' items, respectively. $X_{i,t}$ includes the same control variables as the main regression model: parent age, parent age squared, parent gender, indicator variables for the number of children in the household, and year-month fixed effects. The coefficients a_1 through a_4, conventionally termed 'a-paths', represent the estimated differences between girls' and boys' monthly spending in each category, holding the control variables constant.

The fifth equation models total monthly income as a function of child gender, spending in four expense categories, and the control variables:

$$Y_{i,t} = \alpha_5 + c'Female_i + b_1M_{1,i,t} + b_2M_{2,i,t} + b_3M_{3,i,t} + b_4M_{4,i,t} + X_{i,t}\lambda_5 + \varepsilon_{5,i,t}$$

Here, $Y_{i,t}$ is total monthly income for child i in month t, measured in pounds sterling. The coefficients b_1 through b_4, conventionally termed 'b-paths', describe the difference in predicted monthly income associated with an additional £1 of spending in each category, holding gender, spending in the other three categories, and the control variables constant. The income equation contains no gender-by-spending interactions, so each spending–income association is assumed to be the same for girls and boys within an age-specific model.

Estimation. We estimate each age-specific model by maximum likelihood under Gaussian distributions. The model assumes spherical errors, with residual covariances between equations fixed at zero. Standard errors are clustered at the household level to account for repeated observations of each child and correlation among siblings. From each estimated model, we calculate direct, indirect and total effects as follows:

Direct effect. The direct effect c' is the estimated difference in monthly income between girls and boys when spending in all four categories and the control variables are held constant. A negative value indicates lower predicted income for girls than boys; a positive value indicates higher predicted income for girls than boys.

Indirect effects. The indirect effect for spending category j is the product of its a-path and b-path:

$$Indirect_j = a_j \times b_j.$$

This combines the gender difference in spending in that category with the category's association with income, as defined above. Each indirect effect is measured in pounds of monthly income. A positive value represents an income difference in the direction of girls receiving more, whereas a negative value represents an income difference in the direction of boys receiving more. These effects can partly offset one another and need not have the same sign as the overall income gap. For example, in our estimates, girls' higher clothing spending and the positive association between clothing spending and income produce a positive clothing indirect effect, even at ages when girls receive less income overall. The total indirect effect is the sum of the four category-specific effects and represents the net income difference associated with girls' and boys' different spending patterns.

Total effect. The total effect is the sum of direct and total indirect effects:

$$Total\ Effect = c' + \Sigma_j\ (a_j \times b_j)$$

It represents the overall gender difference in monthly income, holding the control variables constant but not controlling for spending. For these linear models, this equals the gender coefficient from an age-specific regression of income on gender and the same control variables, using the same observations but excluding the spending variables. Thus, the total indirect effect is also the difference between the estimated gender income gap before and after controlling for spending.

Inference. We calculate standard errors for indirect effects using the delta method, which approximates the uncertainty in a product of estimated coefficients. For an indirect effect $a_j b_j$, the estimated variance is:

$$Var(a_j b_j) \approx b_j^2 Var(a_j) + a_j^2 Var(b_j) + 2a_j b_j Cov(a_j, b_j)$$

We implement this calculation using Stata's *nlcom* command and the full household-clustered variance-covariance matrix from the SEM estimation. This approach accounts for covariance between the estimated coefficients when calculating uncertainty in their products. Statistical significance is assessed using two-tailed z-tests, with 95% confidence intervals calculated as the point estimate ± 1.96 × its standard error. All hypothesis tests use a significance threshold of $\alpha = 0.05$.

Data availability statement. The anonymized transaction data are proprietary and were provided by a UK financial services firm under an agreement that does not permit public release. Access for editors, referees or other researchers is subject to the provider's permission.

Code availability statement. Data were prepared in R and statistical analyses were conducted in Stata 19. All R and Stata code and associated output files will be available to editors and referees during peer review and from the authors on request.

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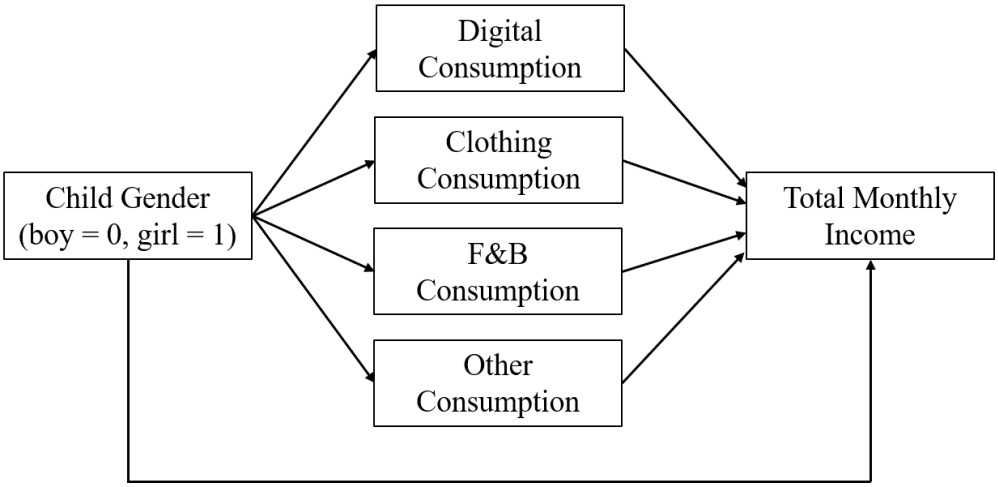
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Author contributions. C.E., R.C.H., M.F.L. and D.P. jointly conceived the study and developed the research design and analytical strategy. All authors contributed to the literature review, development and review of spending classifications, evaluation and interpretation of the results, preparation of figures and tables, and project coordination. M.F.L. secured access to the data and coordinated the relationship with the data provider. Owing to data-access restrictions, direct access to the transaction data was limited to C.E. and M.F.L., who curated and processed the data, constructed the panel dataset, developed and implemented the analysis code, and performed the statistical analyses, with methodological input and review from all authors. All authors contributed to drafting and revising the manuscript.

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Extended Data



Extended Data Fig. 1 | Mediation model. Consumption in each category is measured as a child's total monthly spending in that category. Digital, Clothing, and F&B spending account for over 50% of total spending at most ages. 'Other' accounts for all other spending. See Methods, 'Mediation analysis' for further details.

Extended Data Table 1 | Sample size and gender split at each age

Age	N	Female	%	Male	%
6	108,809	52,622	48.36	56,187	51.64
7	186,646	89,918	48.18	96,728	51.82
8	265,938	130,141	48.94	135,797	51.06
9	346,102	172,888	49.95	173,214	50.05
10	420,960	214,588	50.98	206,372	49.02
11	467,652	242,144	51.78	225,508	48.22
12	453,472	235,659	51.97	217,813	48.03
13	396,974	205,293	51.71	191,681	48.29
14	318,244	163,012	51.22	155,232	48.78
15	234,084	118,320	50.55	115,764	49.45

This table reports the sample size and gender split for each age 6 through 15. The total N in this table exceeds our sample size of unique children (1,154,275; 50.51% female) because some children appear in the data over an extended period. For example, a 10-year-old who has been using the app since she was 6 would be included in the sample of 6-year-olds until her 7th birthday, in the sample of 7-year-olds until her 8th birthday, and so on.

Extended Data Table 2a | The gender income gap

	Monthly income				Cumulative annual income			
Age	Girls	Boys	Gap (£)	Gap (%)	Girls	Boys	Gap (£)	Gap (%)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Childhood (6-10):								
6	22.47	24.00	-1.53***	-6.36***	269.68	287.99	-18.31***	-6.36***
	(0.20)	(0.20)	(0.28)	(1.12)	(2.43)	(2.43)	(3.33)	(1.12)
7	20.52	22.53	-2.01***	-8.94***	515.87	558.35	-42.48***	-7.61***
	(0.14)	(0.15)	(0.20)	(0.83)	(3.44)	(3.48)	(4.63)	(0.80)
8	21.95	24.25	-2.30***	-9.48***	779.32	849.38	-70.06***	-8.25***
	(0.13)	(0.12)	(0.17)	(0.65)	(4.18)	(4.23)	(5.44)	(0.61)
9	23.98	26.94	-2.96***	-10.97***	1067.11	1172.64	-105.53***	-9.00***
	(0.11)	(0.12)	(0.15)	(0.53)	(4.80)	(4.87)	(6.07)	(0.49)
10	27.23	30.33	-3.11***	-10.24***	1393.83	1536.65	-142.81***	-9.29***
	(0.10)	(0.11)	(0.14)	(0.43)	(5.33)	(5.47)	(6.58)	(0.41)
Adolescence (11-15):								
11	33.78	35.01	-1.22***	-3.50***	1799.21	1956.72	-157.51***	-8.05***
	(0.11)	(0.12)	(0.15)	(0.41)	(5.86)	(6.07)	(7.06)	(0.35)
12	43.99	42.13	1.86***	4.41***	2327.04	2462.26	-135.21***	-5.49***
	(0.13)	(0.13)	(0.17)	(0.42)	(6.52)	(6.71)	(7.66)	(0.30)
13	54.97	51.05	3.92***	7.68***	2986.67	3074.81	-88.14***	-2.87***
	(0.16)	(0.16)	(0.22)	(0.44)	(7.37)	(7.49)	(8.51)	(0.27)
14	65.22	60.36	4.86***	8.05***	3769.35	3799.18	-29.84***	-0.79***
	(0.21)	(0.21)	(0.29)	(0.50)	(8.50)	(8.55)	(9.81)	(0.26)
15	74.75	70.21	4.55***	6.48***	4666.40	4641.66	24.74**	0.53**
	(0.29)	(0.28)	(0.40)	(0.58)	(10.17)	(10.07)	(11.86)	(0.26)

This table reports income estimates for girls and boys aged 6 to 15, as measured by our main regression analysis. Columns (1) and (2) report mean monthly income for girls and boys, column (3) reports the difference in monthly income between girls and boys in pounds sterling, and column (4) reports the percentage difference. Columns (5) to (8) present the same information as columns (1) to (4), but for cumulative annual income. Annual income at each age is computed as 12 × the predicted monthly mean for that age (with all covariates set to their sample means); cumulative annual income is the sum of these annual amounts from age 6 to the age in each row. Standard errors are reported in parentheses. ***, **, and * indicate statistical significance from coefficient difference tests at the 1%, 5%, and 10% levels, respectively.

Extended Data Table 2b | The monthly gender income gap split by parent gender

	Panel A: App administrator is mother				Panel B: App administrator is father			
Age	Girls	Boys	Gap (£)	Gap (%)	Girls	Boys	Gap (£)	Gap (%)
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Childhood (6-10):								
6	22.29	23.64	-1.34***	-5.68***	23.06	25.16	-2.10***	-8.36***
	(0.24)	(0.23)	(0.33)	(1.36)	(0.33)	(0.39)	(0.50)	(1.91)
7	20.28	22.41	-2.14***	-9.54***	21.27	22.96	-1.69***	-7.34***
	(0.17)	(0.18)	(0.24)	(1.00)	(0.24)	(0.25)	(0.34)	(1.44)
8	21.66	24.06	-2.40***	-9.98***	22.86	24.88	-2.02***	-8.12***
	(0.15)	(0.14)	(0.20)	(0.79)	(0.21)	(0.22)	(0.30)	(1.15)
9	23.68	26.63	-2.94***	-11.06***	24.89	27.89	-3.00***	-10.75***
	(0.13)	(0.13)	(0.18)	(0.63)	(0.19)	(0.22)	(0.28)	(0.94)
10	26.90	29.90	-3.00***	-10.03***	28.19	31.61	-3.42***	-10.81***
	(0.12)	(0.13)	(0.16)	(0.52)	(0.17)	(0.21)	(0.26)	(0.78)
Adolescence (11-15):								
11	33.43	34.58	-1.15***	-3.32***	34.79	36.23	-1.44***	-3.97***
	(0.13)	(0.13)	(0.17)	(0.49)	(0.19)	(0.21)	(0.27)	(0.74)
12	43.50	41.67	1.83***	4.40***	45.31	43.38	1.93***	4.44***
	(0.15)	(0.15)	(0.20)	(0.50)	(0.23)	(0.24)	(0.32)	(0.76)
13	54.32	50.44	3.88***	7.69***	56.66	52.62	4.05***	7.69***
	(0.19)	(0.19)	(0.26)	(0.54)	(0.30)	(0.29)	(0.40)	(0.79)
14	63.97	59.79	4.18***	6.99***	68.30	61.79	6.51***	10.53***
	(0.25)	(0.25)	(0.34)	(0.59)	(0.40)	(0.38)	(0.53)	(0.91)
15	73.20	69.55	3.66***	5.26***	78.42	71.79	6.63***	9.24***
	(0.35)	(0.34)	(0.47)	(0.69)	(0.55)	(0.52)	(0.73)	(1.07)

This table reports monthly income estimates for girls and boys aged 6 to 15 split by parent gender, as measured by our main regression analysis. Panel A reports the results when the app is administered by a child's mother, and Panel B reports the results when the app is administered by a child's father. In each panel, columns (1) and (2) report mean monthly income for girls and boys, column (3) reports the difference in monthly income between girls and boys in pounds sterling, and column (4) reports the percentage difference. Standard errors are reported in parentheses. ***, **, and * indicate statistical significance from coefficient difference tests at the 1%, 5%, and 10% levels, respectively.

Extended Data Table 2c | The gender income gap for girl–boy siblings

	Monthly income				Cumulative annual income			
Age	Girls	Boys	Gap (£)	Gap (%)	Girls	Boys	Gap (£)	Gap (%)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Childhood (6-10):								
6	23.61	24.58	-0.97*	-3.95**	283.29	294.95	-11.66*	-3.95**
	(0.70)	(0.70)	(0.50)	(1.98)	(8.40)	(8.45)	(5.96)	(1.98)
7	19.98	21.52	-1.54***	-7.17***	523.01	553.17	-30.16***	-5.45***
	(0.57)	(0.56)	(0.35)	(1.56)	(14.65)	(14.63)	(8.40)	(1.48)
8	19.56	21.31	-1.75***	-8.20***	757.78	808.92	-51.14***	-6.32***
	(0.48)	(0.47)	(0.29)	(1.32)	(19.68)	(19.56)	(9.68)	(1.17)
9	19.64	22.01	-2.37***	-10.77***	993.45	1073.04	-79.59***	-7.42***
	(0.42)	(0.42)	(0.25)	(1.11)	(23.74)	(23.56)	(10.09)	(0.92)
10	21.16	23.54	-2.37***	-10.09***	1247.43	1355.51	-108.08***	-7.97***
	(0.38)	(0.39)	(0.25)	(1.01)	(27.00)	(26.88)	(10.06)	(0.73)
Adolescence (11-15):								
11	25.74	26.54	-0.80***	-3.03***	1556.30	1674.03	-117.72***	-7.03***
	(0.38)	(0.38)	(0.26)	(0.97)	(29.61)	(29.52)	(10.03)	(0.59)
12	34.00	32.07	1.93***	6.03***	1964.32	2058.86	-94.54***	-4.59***
	(0.40)	(0.40)	(0.30)	(0.96)	(31.75)	(31.69)	(10.32)	(0.50)
13	44.41	39.75	4.65***	11.71***	2497.20	2535.88	-38.68***	-1.53***
	(0.45)	(0.44)	(0.37)	(0.99)	(33.66)	(33.63)	(11.30)	(0.44)
14	52.56	47.83	4.73***	9.89***	3127.97	3109.88	18.09	0.58
	(0.53)	(0.51)	(0.50)	(1.09)	(35.59)	(35.49)	(13.14)	(0.42)
15	60.66	56.37	4.29***	7.61***	3855.89	3786.32	69.57***	1.84***
	(0.66)	(0.67)	(0.72)	(1.33)	(37.78)	(37.49)	(16.68)	(0.44)

This table reports income estimates for the 135,159 pairs of girl–boy siblings in the data. Results were obtained using our main regression analysis, but with the addition of a household fixed effect, so that gender differences are estimated by comparing girls and boys in the same household. The sample comprises household-by-month observations with exactly two enrolled children, one girl and one boy, and the household fixed effect replaces the parent-gender and number-of-children controls used in the main model; the child age-by-gender interaction, parent age and its square, year-month fixed effects, and household-clustered standard errors are otherwise unchanged. Columns (1) and (2) report mean monthly income for girls and boys, column (3) reports the difference in monthly income between girls and boys in pounds sterling, and column (4) reports the percentage difference. Columns (5) to (8) present the same information as columns (1) to (4), but for cumulative annual income. Annual income at each age is computed as 12 × the predicted monthly mean for that age (with all covariates set to their sample means); cumulative annual income is the sum of these annual amounts from age 6 to the age in each row. Standard errors are reported in parentheses. ***, **, and * indicate statistical significance from coefficient difference tests at the 1%, 5%, and 10% levels, respectively.

Extended Data Table 3 | The gender savings gap

	Monthly savings				Cumulative annual savings			
Age	Girls	Boys	Gap (£)	Gap (%)	Girls	Boys	Gap (£)	Gap (%)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Childhood (6-10):								
6	11.26	12.23	-0.97***	-7.96***	135.08	146.76	-11.69***	-7.96***
	(0.12)	(0.12)	(0.16)	(1.27)	(1.43)	(1.38)	(1.94)	(1.27)
7	8.85	10.00	-1.16***	-11.56***	241.24	266.80	-25.57***	-9.58***
	(0.08)	(0.08)	(0.11)	(1.02)	(1.89)	(1.87)	(2.55)	(0.91)
8	9.23	10.31	-1.08***	-10.48***	351.94	390.47	-38.52***	-9.87***
	(0.07)	(0.07)	(0.09)	(0.84)	(2.20)	(2.20)	(2.90)	(0.71)
9	9.68	11.03	-1.35***	-12.21***	468.16	522.85	-54.69***	-10.46***
	(0.06)	(0.06)	(0.08)	(0.70)	(2.47)	(2.47)	(3.16)	(0.57)
10	10.11	11.74	-1.63***	-13.87***	589.49	663.72	-74.22***	-11.18***
	(0.05)	(0.06)	(0.08)	(0.60)	(2.69)	(2.73)	(3.37)	(0.48)
Adolescence (11-15):								
11	11.05	12.51	-1.47***	-11.72***	722.05	813.86	-91.81***	-11.28***
	(0.05)	(0.06)	(0.08)	(0.58)	(2.90)	(2.99)	(3.56)	(0.41)
12	13.42	14.06	-0.64***	-4.53***	883.13	982.59	-99.46***	-10.12***
	(0.06)	(0.07)	(0.09)	(0.60)	(3.15)	(3.26)	(3.80)	(0.37)
13	15.67	15.74	-0.08	-0.49	1071.15	1171.52	-100.38***	-8.57***
	(0.08)	(0.08)	(0.11)	(0.68)	(3.46)	(3.59)	(4.13)	(0.34)
14	17.27	17.40	-0.12	-0.71	1278.41	1380.28	-101.87***	-7.38***
	(0.10)	(0.10)	(0.14)	(0.80)	(3.88)	(4.02)	(4.62)	(0.32)
15	18.58	18.73	-0.15	-0.82	1501.35	1605.06	-103.71***	-6.46***
	(0.14)	(0.14)	(0.19)	(1.03)	(4.53)	(4.64)	(5.43)	(0.33)

This table reports savings estimates for girls and boys aged 6 to 15, as measured using our main regression analysis with savings as the dependent variable. Columns (1) and (2) report mean monthly savings for girls and boys, column (3) reports the difference in monthly savings between girls and boys in pounds sterling, and column (4) reports the percentage difference. Columns (5) to (8) present the same information as columns (1) to (4), but for cumulative annual savings. The annual savings estimate at each age is computed as 12 × the predicted monthly mean for that age (with all covariates set to their sample means); cumulative annual savings estimates are the sum of these annual amounts from age 6 to the age in each row. Standard errors are reported in parentheses. ***, **, and * indicate statistical significance from coefficient difference tests at the 1%, 5%, and 10% levels, respectively.

Extended Data Table 4a | The gender gap for total spending

	Monthly spending				Cumulative annual spending			
Age	Girls	Boys	Gap (£)	Gap (%)	Girls	Boys	Gap (£)	Gap (%)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Childhood (6-10):								
6	11.22	11.77	-0.55***	-4.69***	134.61	141.23	-6.62***	-4.69***
	(0.12)	(0.13)	(0.17)	(1.44)	(1.46)	(1.58)	(2.08)	(1.44)
7	11.67	12.53	-0.86***	-6.85***	274.63	291.55	-16.91***	-5.80***
	(0.10)	(0.10)	(0.13)	(1.00)	(2.18)	(2.29)	(2.98)	(0.99)
8	12.73	13.95	-1.22***	-8.74***	427.38	458.91	-31.54***	-6.87***
	(0.08)	(0.08)	(0.11)	(0.75)	(2.70)	(2.82)	(3.56)	(0.75)
9	14.30	15.91	-1.61***	-10.11***	598.95	649.79	-50.84***	-7.82***
	(0.07)	(0.08)	(0.10)	(0.59)	(3.12)	(3.28)	(4.02)	(0.59)
10	17.12	18.59	-1.48***	-7.95***	804.34	872.93	-68.59***	-7.86***
	(0.07)	(0.08)	(0.09)	(0.47)	(3.48)	(3.71)	(4.39)	(0.48)
Adolescence (11-15):								
11	22.73	22.49	0.24**	1.07**	1077.16	1142.85	-65.70***	-5.75***
	(0.07)	(0.08)	(0.10)	(0.43)	(3.86)	(4.13)	(4.73)	(0.40)
12	30.56	28.07	2.50***	8.89***	1443.91	1479.66	-35.76***	-2.42***
	(0.09)	(0.09)	(0.12)	(0.43)	(4.32)	(4.59)	(5.16)	(0.34)
13	39.30	35.30	4.00***	11.33***	1915.52	1903.29	12.24**	0.64**
	(0.11)	(0.11)	(0.15)	(0.44)	(4.93)	(5.11)	(5.76)	(0.30)
14	47.95	42.97	4.98***	11.60***	2490.94	2418.91	72.03***	2.98***
	(0.15)	(0.14)	(0.20)	(0.48)	(5.75)	(5.83)	(6.69)	(0.28)
15	56.18	51.47	4.70***	9.13***	3165.05	3036.60	128.45***	4.23***
	(0.21)	(0.20)	(0.28)	(0.56)	(6.98)	(6.92)	(8.21)	(0.28)

This table reports spending estimates for girls and boys aged 6 to 15, as measured using our main regression analysis with total spending across all categories as the dependent variable. Columns (1) and (2) report mean monthly spending for girls and boys, column (3) reports the difference in monthly spending between girls and boys in pounds sterling, and column (4) reports the percentage difference. Columns (5) to (8) present the same information as columns (1) to (4), but for cumulative annual spending. Annual spending at each age is computed as 12 × the predicted monthly mean for that age (with all covariates set to their sample means); cumulative annual spending is the sum of these annual amounts from age 6 to the age in each row. Standard errors are reported in parentheses. ***, **, and * indicate statistical significance from coefficient difference tests at the 1%, 5%, and 10% levels, respectively.

Extended Data Table 4b | Gender differences in average and marginal propensity to consume

	Average Propensity to Consume			Marginal Propensity to Consume		
Age	Girls	Boys	Gap	Girls	Boys	Gap
Childhood (6-10):						
6	0.499	0.490	0.009	0.366	0.352	0.014
				(0.021)	(0.015)	(0.025)
7	0.569	0.556	0.013	0.386	0.359	0.027*
				(0.010)	(0.011)	(0.015)
8	0.580	0.575	0.005	0.356	0.335	0.021**
				(0.008)	(0.006)	(0.010)
9	0.596	0.591	0.006	0.344	0.331	0.013
				(0.006)	(0.006)	(0.008)
10	0.629	0.613	0.016	0.331	0.326	0.005
				(0.008)	(0.005)	(0.009)
Adolescence (11-15):						
11	0.673	0.643	0.030	0.353	0.332	0.021***
				(0.005)	(0.005)	(0.007)
12	0.695	0.666	0.029	0.382	0.344	0.038***
				(0.004)	(0.004)	(0.005)
13	0.715	0.692	0.023	0.395	0.363	0.032***
				(0.004)	(0.004)	(0.005)
14	0.735	0.712	0.023	0.405	0.373	0.032***
				(0.004)	(0.004)	(0.006)
15	0.752	0.733	0.018	0.406	0.385	0.021**
				(0.006)	(0.005)	(0.008)

This table reports the average propensity to consume (APC) and the marginal propensity to consume (MPC) for girls and boys aged 6 to 15. APC is purely descriptive: we calculate it as estimated total monthly spending (Extended Data Table 4a) divided by estimated total monthly income (Extended Data Table 2a). APC estimates and the corresponding gender differences are reported descriptively, without hypothesis tests or standard errors. MPC is estimated from a single regression of total monthly spending on total monthly income, with income interacted with child age and child gender, and the same control variables and household-clustered standard errors as our main regression. The MPC in each cell is the marginal effect of income on spending for that age and gender; the MPC gaps are the girl-minus-boy differences in this marginal effect at each age. For MPC, standard errors are reported in parentheses and ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Extended Data Table 5 | The gender income gap in each income category

	Monthly income				Cumulative annual income			
Age	Girls	Boys	Gap (£)	Gap (%)	Girls	Boys	Gap (£)	Gap (%)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Ad hoc transfers								
Childhood (6-10):								
6	9.62	10.72	-1.10***	-10.28***	115.40	128.62	-13.22***	-10.28***
	(0.12)	(0.13)	(0.17)	(1.49)	(1.40)	(1.55)	(2.03)	(1.49)
7	8.43	9.63	-1.20***	-12.48***	216.53	244.18	-27.65***	-11.32***
	(0.09)	(0.09)	(0.12)	(1.18)	(2.01)	(2.17)	(2.80)	(1.08)
8	8.93	10.40	-1.48***	-14.19***	323.66	369.02	-45.36***	-12.29***
	(0.08)	(0.07)	(0.10)	(0.89)	(2.47)	(2.60)	(3.29)	(0.84)
9	9.75	11.60	-1.85***	-15.97***	440.60	508.18	-67.58***	-13.30***
	(0.07)	(0.07)	(0.09)	(0.69)	(2.84)	(2.95)	(3.65)	(0.67)
10	11.20	13.28	-2.08***	-15.63***	575.05	667.54	-92.49***	-13.85***
	(0.06)	(0.07)	(0.08)	(0.56)	(3.14)	(3.29)	(3.94)	(0.55)
Adolescence (11-15):								
11	14.47	15.51	-1.04***	-6.70***	748.66	853.62	-104.95***	-12.30***
	(0.06)	(0.07)	(0.08)	(0.53)	(3.45)	(3.62)	(4.21)	(0.46)
12	19.81	19.24	0.57***	2.95***	986.33	1084.46	-98.13***	-9.05***
	(0.07)	(0.08)	(0.10)	(0.52)	(3.81)	(3.99)	(4.55)	(0.40)
13	25.70	24.09	1.61***	6.68***	1294.75	1373.57	-78.82***	-5.74***
	(0.09)	(0.09)	(0.12)	(0.53)	(4.29)	(4.44)	(5.04)	(0.36)
14	31.51	29.37	2.15***	7.31***	1672.90	1725.97	-53.07***	-3.07***
	(0.12)	(0.12)	(0.17)	(0.59)	(4.95)	(5.06)	(5.80)	(0.33)
15	36.79	34.94	1.86***	5.31***	2114.40	2145.20	-30.80***	-1.44***
	(0.17)	(0.16)	(0.23)	(0.66)	(5.90)	(5.97)	(6.99)	(0.32)
Panel B: Monetary gifts								
Childhood (6-10):								
6	5.87	6.25	-0.38***	-6.14***	70.41	75.01	-4.61***	-6.14***
	(0.10)	(0.09)	(0.13)	(2.03)	(1.18)	(1.10)	(1.57)	(2.03)
7	5.22	6.09	-0.87***	-14.34***	133.02	148.11	-15.09***	-10.19***
	(0.07)	(0.07)	(0.09)	(1.37)	(1.59)	(1.55)	(2.11)	(1.36)
8	5.81	6.70	-0.89***	-13.32***	202.74	228.54	-25.80***	-11.29***
	(0.06)	(0.06)	(0.08)	(1.07)	(1.87)	(1.85)	(2.43)	(1.01)
9	6.52	7.56	-1.05***	-13.83***	280.96	319.32	-38.36***	-12.01***
	(0.05)	(0.05)	(0.07)	(0.86)	(2.12)	(2.09)	(2.67)	(0.79)
10	7.56	8.59	-1.04***	-12.05***	371.66	422.43	-50.78***	-12.02***
	(0.05)	(0.05)	(0.07)	(0.71)	(2.33)	(2.33)	(2.87)	(0.64)
Adolescence (11-15):								
11	9.69	10.05	-0.35***	-3.52***	487.96	542.98	-55.02***	-10.13***
	(0.05)	(0.05)	(0.07)	(0.67)	(2.55)	(2.57)	(3.07)	(0.54)
12	13.08	12.26	0.82***	6.73***	644.95	690.07	-45.12***	-6.54***
	(0.06)	(0.06)	(0.08)	(0.68)	(2.81)	(2.83)	(3.32)	(0.47)
13	16.71	15.03	1.69***	11.23***	845.51	870.38	-24.87***	-2.86***
	(0.08)	(0.07)	(0.10)	(0.72)	(3.15)	(3.16)	(3.68)	(0.42)
14	19.63	17.56	2.07***	11.81***	1081.06	1081.04	0.01	0.00
	(0.10)	(0.09)	(0.13)	(0.80)	(3.60)	(3.60)	(4.22)	(0.39)
15	22.45	20.41	2.04***	9.99***	1350.47	1325.98	24.49***	1.85***
	(0.13)	(0.13)	(0.18)	(0.93)	(4.26)	(4.21)	(5.05)	(0.38)

Extended Data Table 5 (continued)

Panel C: Allowances								
Childhood (6-10):								
6	6.43	6.47	-0.05	-0.72	77.14	77.70	-0.56	-0.72
	(0.06)	(0.05)	(0.08)	(1.16)	(0.69)	(0.65)	(0.91)	(1.16)
7	6.37	6.32	0.05	0.80	153.59	153.54	0.05	0.03
	(0.04)	(0.04)	(0.05)	(0.87)	(1.08)	(1.01)	(1.38)	(0.90)
8	6.67	6.61	0.06	0.85	233.62	232.89	0.73	0.31
	(0.04)	(0.04)	(0.05)	(0.72)	(1.34)	(1.30)	(1.69)	(0.73)
9	7.13	7.20	-0.07	-1.03	319.16	319.32	-0.17	-0.05
	(0.03)	(0.04)	(0.05)	(0.65)	(1.55)	(1.59)	(1.96)	(0.61)
10	7.82	7.86	-0.04	-0.54	412.98	413.65	-0.68	-0.16
	(0.03)	(0.04)	(0.04)	(0.54)	(1.73)	(1.86)	(2.17)	(0.52)
Adolescence (11-15):								
11	8.92	8.82	0.11**	1.19**	520.06	519.47	0.59	0.11
	(0.03)	(0.03)	(0.04)	(0.48)	(1.92)	(2.10)	(2.37)	(0.46)
12	10.33	9.95	0.38***	3.87***	644.02	638.82	5.20**	0.81**
	(0.04)	(0.04)	(0.05)	(0.49)	(2.15)	(2.34)	(2.58)	(0.41)
13	11.75	11.21	0.53***	4.77***	785.00	773.39	11.62***	1.50***
	(0.05)	(0.04)	(0.06)	(0.53)	(2.44)	(2.56)	(2.80)	(0.36)
14	13.28	12.73	0.55***	4.31***	944.39	926.19	18.20***	1.97***
	(0.06)	(0.06)	(0.07)	(0.59)	(2.77)	(2.85)	(3.13)	(0.34)
15	14.76	14.18	0.59***	4.14***	1121.53	1096.29	25.24***	2.30***
	(0.08)	(0.07)	(0.10)	(0.72)	(3.24)	(3.27)	(3.64)	(0.34)
Panel D: Task payments								
Childhood (6-10):								
6	0.56	0.55	0.01	1.17	6.74	6.66	0.08	1.17
	(0.01)	(0.01)	(0.01)	(2.25)	(0.11)	(0.11)	(0.15)	(2.25)
7	0.50	0.49	0.01	2.10	12.73	12.53	0.20	1.61
	(0.01)	(0.01)	(0.01)	(1.83)	(0.17)	(0.16)	(0.22)	(1.73)
8	0.55	0.53	0.01*	2.70*	19.30	18.93	0.37	1.98
	(0.01)	(0.01)	(0.01)	(1.59)	(0.21)	(0.20)	(0.26)	(1.39)
9	0.59	0.57	0.02**	2.93**	26.39	25.81	0.58*	2.23*
	(0.01)	(0.01)	(0.01)	(1.37)	(0.24)	(0.24)	(0.30)	(1.17)
10	0.65	0.60	0.05***	7.62***	34.15	33.02	1.12***	3.41***
	(0.01)	(0.01)	(0.01)	(1.29)	(0.28)	(0.28)	(0.33)	(1.02)
Adolescence (11-15):								
11	0.70	0.64	0.06***	9.84***	42.53	40.65	1.88***	4.61***
	(0.01)	(0.01)	(0.01)	(1.29)	(0.31)	(0.31)	(0.36)	(0.91)
12	0.77	0.69	0.08***	11.71***	51.74	48.90	2.84***	5.81***
	(0.01)	(0.01)	(0.01)	(1.42)	(0.35)	(0.35)	(0.40)	(0.84)
13	0.81	0.71	0.09***	12.71***	61.41	57.47	3.93***	6.84***
	(0.01)	(0.01)	(0.01)	(1.66)	(0.40)	(0.39)	(0.45)	(0.81)
14	0.80	0.71	0.09***	12.76***	71.00	65.98	5.02***	7.60***
	(0.01)	(0.01)	(0.01)	(2.07)	(0.45)	(0.45)	(0.51)	(0.81)
15	0.75	0.68	0.07***	9.58***	80.00	74.20	5.80***	7.82***
	(0.01)	(0.01)	(0.02)	(2.60)	(0.53)	(0.52)	(0.60)	(0.85)

Extended Data Table 5 (notes)

This table reports estimated monthly income from each income stream using our main regression analysis. Panel A is ad hoc transfer income, Panel B is monetary gift income, Panel C is allowance income, and Panel D is task income. Columns (1) and (2) report mean monthly income for girls and boys, column (3) reports the difference in monthly income between girls and boys in pounds sterling, and column (4) reports the percentage difference. Columns (5) to (8) present the same information as columns (1) to (4), but for cumulative annual income. Annual income at each age is computed as 12 × the predicted monthly mean for that age (with all covariates set to their sample means); cumulative annual income is the sum of these annual amounts from age 6 to the age in each row. Standard errors are reported in parentheses. ***, **, and * indicate statistical significance from coefficient difference tests at the 1%, 5%, and 10% levels, respectively.

Extended Data Table 6 | The number and amount of income transfers in each category

	Average # of transfers per month				Average amount per transfer			
Age	Girls	Boys	Gap (#)	Gap (%)	Girls	Boys	Gap (£)	Gap (%)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Ad hoc transfers								
Childhood (6-10):								
6	0.75	0.81	-0.06***	-7.87***	13.01	13.30	-0.30*	-2.22*
	(0.01)	(0.01)	(0.01)	(0.90)	(0.11)	(0.11)	(0.16)	(1.17)
7	0.60	0.69	-0.10***	-14.14***	14.18	13.96	0.22	1.58
	(0.00)	(0.00)	(0.01)	(0.74)	(0.10)	(0.09)	(0.14)	(0.99)
8	0.61	0.75	-0.13***	-17.66***	14.56	14.02	0.55***	3.91***
	(0.00)	(0.00)	(0.00)	(0.60)	(0.09)	(0.07)	(0.11)	(0.79)
9	0.66	0.84	-0.18***	-21.14***	14.78	13.89	0.89***	6.42***
	(0.00)	(0.00)	(0.00)	(0.48)	(0.07)	(0.06)	(0.09)	(0.63)
10	0.76	0.95	-0.20***	-20.58***	14.89	14.00	0.89***	6.39***
	(0.00)	(0.00)	(0.00)	(0.42)	(0.06)	(0.05)	(0.07)	(0.52)
Adolescence (11-15):								
11	1.00	1.13	-0.13***	-11.68***	14.63	13.80	0.83***	6.04***
	(0.00)	(0.00)	(0.01)	(0.43)	(0.05)	(0.04)	(0.06)	(0.44)
12	1.32	1.40	-0.08***	-5.79***	15.15	13.83	1.31***	9.49***
	(0.00)	(0.01)	(0.01)	(0.45)	(0.04)	(0.04)	(0.06)	(0.42)
13	1.57	1.65	-0.08***	-4.99***	16.42	14.63	1.79***	12.20***
	(0.01)	(0.01)	(0.01)	(0.48)	(0.05)	(0.04)	(0.06)	(0.43)
14	1.76	1.85	-0.10***	-5.13***	17.97	15.95	2.03***	12.72***
	(0.01)	(0.01)	(0.01)	(0.52)	(0.05)	(0.05)	(0.07)	(0.48)
15	1.87	1.99	-0.12***	-6.02***	19.69	17.63	2.06***	11.66***
	(0.01)	(0.01)	(0.01)	(0.61)	(0.07)	(0.07)	(0.09)	(0.56)
Panel B: Gifts								
Childhood (6-10):								
6	0.38	0.41	-0.03***	-6.75***	15.25	15.18	0.08	0.52
	(0.00)	(0.00)	(0.01)	(1.43)	(0.18)	(0.15)	(0.23)	(1.52)
7	0.32	0.39	-0.07***	-17.36***	16.06	15.55	0.51***	3.29***
	(0.00)	(0.00)	(0.00)	(1.08)	(0.13)	(0.11)	(0.17)	(1.08)
8	0.35	0.43	-0.08***	-18.34***	16.55	15.71	0.83***	5.30***
	(0.00)	(0.00)	(0.00)	(0.86)	(0.11)	(0.09)	(0.13)	(0.87)
9	0.39	0.48	-0.09***	-19.16***	16.58	15.63	0.95***	6.06***
	(0.00)	(0.00)	(0.00)	(0.69)	(0.09)	(0.07)	(0.11)	(0.72)
10	0.46	0.55	-0.09***	-16.56***	16.49	15.64	0.84***	5.40***
	(0.00)	(0.00)	(0.00)	(0.60)	(0.07)	(0.06)	(0.09)	(0.59)
Adolescence (11-15):								
11	0.60	0.66	-0.05***	-8.13***	16.14	15.33	0.80***	5.24***
	(0.00)	(0.00)	(0.00)	(0.59)	(0.06)	(0.06)	(0.08)	(0.52)
12	0.81	0.82	-0.01***	-1.67***	16.33	15.00	1.32***	8.81***
	(0.00)	(0.00)	(0.01)	(0.62)	(0.05)	(0.05)	(0.07)	(0.51)
13	0.98	0.98	0.00	0.31	17.12	15.45	1.67***	10.84***
	(0.00)	(0.00)	(0.01)	(0.67)	(0.06)	(0.06)	(0.08)	(0.53)
14	1.07	1.07	-0.00	-0.10	18.45	16.55	1.90***	11.48***
	(0.01)	(0.01)	(0.01)	(0.76)	(0.07)	(0.07)	(0.09)	(0.59)
15	1.12	1.13	-0.01	-0.97	20.17	18.25	1.92***	10.51***
	(0.01)	(0.01)	(0.01)	(0.89)	(0.09)	(0.09)	(0.12)	(0.71)

Extended Data Table 6 (continued)

Panel C: Allowances								
Childhood (6-10):								
6	2.11	2.12	-0.00	-0.19	3.11	3.13	-0.02	-0.70
	(0.01)	(0.01)	(0.01)	(0.50)	(0.02)	(0.02)	(0.03)	(1.02)
7	2.15	2.14	0.01	0.52	3.02	3.02	0.00	0.04
	(0.01)	(0.01)	(0.01)	(0.38)	(0.02)	(0.02)	(0.02)	(0.73)
8	2.20	2.19	0.01*	0.61*	3.08	3.08	0.00	0.04
	(0.01)	(0.01)	(0.01)	(0.31)	(0.01)	(0.01)	(0.02)	(0.61)
9	2.25	2.26	-0.01	-0.32	3.23	3.26	-0.03	-0.89
	(0.00)	(0.00)	(0.01)	(0.26)	(0.01)	(0.02)	(0.02)	(0.57)
10	2.30	2.30	-0.00	-0.12	3.46	3.48	-0.02	-0.64
	(0.00)	(0.00)	(0.01)	(0.23)	(0.01)	(0.01)	(0.02)	(0.46)
Adolescence (11-15):								
11	2.35	2.35	0.00	0.10	3.87	3.84	0.04**	0.91**
	(0.00)	(0.00)	(0.01)	(0.21)	(0.01)	(0.01)	(0.02)	(0.40)
12	2.40	2.38	0.02***	0.72***	4.39	4.26	0.13***	3.02***
	(0.00)	(0.00)	(0.01)	(0.21)	(0.01)	(0.01)	(0.02)	(0.40)
13	2.43	2.41	0.02***	0.76***	4.94	4.75	0.19***	3.94***
	(0.00)	(0.00)	(0.01)	(0.22)	(0.02)	(0.01)	(0.02)	(0.43)
14	2.44	2.43	0.01	0.40	5.58	5.37	0.21***	3.90***
	(0.00)	(0.00)	(0.01)	(0.25)	(0.02)	(0.02)	(0.03)	(0.48)
15	2.45	2.43	0.01*	0.49*	6.18	5.97	0.21***	3.60***
	(0.01)	(0.01)	(0.01)	(0.28)	(0.03)	(0.03)	(0.03)	(0.59)
Panel D: Tasks								
Childhood (6-10):								
6	0.75	0.74	0.01	1.95	0.74	0.74	-0.01	-1.07
	(0.01)	(0.01)	(0.02)	(2.35)	(0.01)	(0.01)	(0.01)	(1.63)
7	0.64	0.62	0.02*	3.72*	0.77	0.79	-0.01	-1.83
	(0.01)	(0.01)	(0.01)	(1.96)	(0.01)	(0.01)	(0.01)	(1.26)
8	0.68	0.62	0.05***	8.41***	0.80	0.85	-0.05***	-5.60***
	(0.01)	(0.01)	(0.01)	(1.64)	(0.01)	(0.01)	(0.01)	(1.05)
9	0.69	0.64	0.05***	7.04***	0.85	0.89	-0.03***	-3.84***
	(0.01)	(0.01)	(0.01)	(1.43)	(0.01)	(0.01)	(0.01)	(0.90)
10	0.72	0.64	0.07***	11.40***	0.90	0.93	-0.03***	-3.26***
	(0.01)	(0.01)	(0.01)	(1.33)	(0.01)	(0.01)	(0.01)	(0.78)
Adolescence (11-15):								
11	0.72	0.64	0.08***	12.03***	0.98	1.00	-0.02**	-1.95**
	(0.01)	(0.01)	(0.01)	(1.33)	(0.01)	(0.01)	(0.01)	(0.79)
12	0.71	0.63	0.08***	13.14***	1.08	1.10	-0.02*	-1.59*
	(0.01)	(0.01)	(0.01)	(1.41)	(0.01)	(0.01)	(0.01)	(0.88)
13	0.68	0.60	0.07***	11.83***	1.21	1.20	0.01	0.45
	(0.01)	(0.01)	(0.01)	(1.59)	(0.01)	(0.01)	(0.01)	(1.07)
14	0.60	0.54	0.06***	10.67***	1.36	1.34	0.02	1.40
	(0.01)	(0.01)	(0.01)	(1.90)	(0.01)	(0.01)	(0.02)	(1.40)
15	0.51	0.48	0.03***	7.06***	1.51	1.49	0.02	1.65
	(0.01)	(0.01)	(0.01)	(2.34)	(0.02)	(0.02)	(0.03)	(1.80)

Extended Data Table 6 (notes)

This table reports the estimated number of income transfers per month and the average amount per income transfer for ad hoc transfers (Panel A), monetary gifts (Panel B), allowances (Panel C), and tasks (Panel D). Estimates were obtained using our main regression analysis. In each panel, columns (1) and (2) report the mean number of transfers per month for girls and boys (calculated at the child-month level), column (3) reports the mean difference in number of transfers per month, and column (4) reports the percentage difference in number of transfers per month. Columns (5) and (6) report the average £ amount per transfer for girls and boys (calculated at the transaction level), column (7) reports the mean difference in average amount per transfer (in pounds sterling) between girls and boys, and column (8) reports the percentage difference. Standard errors are reported in parentheses. ***, **, and * indicate statistical significance from coefficient difference tests at the 1%, 5%, and 10% levels, respectively.

Extended Data Table 7 | Evidence of gendered consumption norms: girls' and boys' spending differs by category

	Panel A: Clothing and Accessories				Panel B: Digital Goods and Services				Panel C: Food and Beverage				Panel D: Other			
Age	Girls	Boys	Gap (£)	Gap (%)	Girls	Boys	Gap (£)	Gap (%)	Girls	Boys	Gap (£)	Gap (%)	Girls	Boys	Gap (£)	Gap (%)
Childhood (6-10):																
6	1.08	0.71	0.37***	52.60***	0.59	1.96	-1.38***	-70.22***	3.35	3.02	0.33***	10.85***	6.58	6.49	0.10	1.48
	(0.02)	(0.01)	(0.02)	(3.97)	(0.02)	(0.04)	(0.04)	(1.00)	(0.05)	(0.05)	(0.07)	(2.34)	(0.07)	(0.08)	(0.11)	(1.71)
7	1.36	0.79	0.58***	73.67***	0.96	3.19	-2.23***	-69.96***	3.10	2.71	0.39***	14.34***	6.61	6.23	0.39***	6.20***
	(0.01)	(0.01)	(0.02)	(2.91)	(0.02)	(0.04)	(0.04)	(0.61)	(0.03)	(0.03)	(0.05)	(1.83)	(0.06)	(0.06)	(0.08)	(1.36)
8	1.75	0.88	0.87***	99.04***	1.40	4.79	-3.39***	-70.71***	3.16	2.63	0.53***	20.02***	6.76	6.01	0.75***	12.49***
	(0.01)	(0.01)	(0.02)	(2.64)	(0.02)	(0.04)	(0.04)	(0.42)	(0.03)	(0.03)	(0.04)	(1.73)	(0.05)	(0.04)	(0.06)	(1.14)
9	2.26	1.04	1.22***	117.76***	1.86	6.42	-4.56***	-71.06***	3.46	2.77	0.69***	24.73***	7.07	6.02	1.05***	17.41***
	(0.01)	(0.01)	(0.02)	(2.38)	(0.02)	(0.04)	(0.04)	(0.31)	(0.02)	(0.03)	(0.04)	(1.73)	(0.04)	(0.04)	(0.05)	(0.98)
10	3.12	1.27	1.85***	145.43***	2.12	7.94	-5.83***	-73.37***	4.36	3.31	1.06***	31.89***	7.86	6.41	1.46***	22.71***
	(0.01)	(0.01)	(0.02)	(2.25)	(0.02)	(0.04)	(0.04)	(0.24)	(0.02)	(0.03)	(0.04)	(1.33)	(0.03)	(0.04)	(0.05)	(0.85)
Adolescence (11-15):																
11	4.58	1.62	2.96***	182.81***	2.05	8.85	-6.81***	-76.86***	6.78	4.92	1.86***	37.72***	9.68	7.42	2.26***	30.41***
	(0.02)	(0.01)	(0.02)	(2.41)	(0.01)	(0.04)	(0.04)	(0.19)	(0.03)	(0.03)	(0.04)	(0.93)	(0.04)	(0.04)	(0.05)	(0.79)
12	6.96	2.20	4.75***	215.94***	2.03	10.00	-7.97***	-79.70***	9.67	7.11	2.56***	36.04***	12.27	9.07	3.20***	35.35***
	(0.03)	(0.02)	(0.03)	(2.62)	(0.01)	(0.04)	(0.05)	(0.17)	(0.03)	(0.03)	(0.05)	(0.76)	(0.05)	(0.04)	(0.06)	(0.76)
13	9.85	3.23	6.62***	205.07***	2.22	11.16	-8.94***	-80.09***	12.41	9.75	2.65***	27.22***	15.22	11.43	3.79***	33.11***
	(0.04)	(0.02)	(0.05)	(2.54)	(0.02)	(0.05)	(0.05)	(0.17)	(0.04)	(0.04)	(0.06)	(0.67)	(0.05)	(0.05)	(0.07)	(0.72)
14	12.37	4.64	7.73***	166.70***	2.48	11.48	-9.00***	-78.40***	14.88	12.41	2.47***	19.87***	18.63	14.71	3.92***	26.65***
	(0.05)	(0.04)	(0.06)	(2.35)	(0.02)	(0.06)	(0.06)	(0.20)	(0.06)	(0.06)	(0.08)	(0.69)	(0.07)	(0.07)	(0.09)	(0.72)
15	14.39	6.18	8.22***	133.05***	2.77	11.29	-8.52***	-75.49***	16.86	15.08	1.78***	11.78***	22.55	19.22	3.33***	17.30***
	(0.08)	(0.05)	(0.09)	(2.29)	(0.02)	(0.06)	(0.07)	(0.25)	(0.08)	(0.08)	(0.10)	(0.74)	(0.10)	(0.10)	(0.13)	(0.77)

This table reports mean monthly spending in each spending category. In each panel A–D, the "Girls" and "Boys" columns report mean monthly spending for each gender in the relevant category, computed as unadjusted subgroup means by age and gender. The "Gap (£)" column reports the difference in monthly spending between girls and boys in pounds sterling, and the "Gap (%)" column reports the percentage difference relative to the boys' mean. All estimates in each panel (means, gaps, standard errors, and significance tests) are obtained from a single fully interacted regression of category spending on age-by-gender cells, whose coefficients equal the subgroup means, with standard errors clustered at the household level. Standard errors are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Extended Data Table 8 | COVID-19 lockdown results

Panel A: Income

	Girls		Boys		
Category	Estimate	SE	Estimate	SE	*P* value
Total Income	-0.47	0.03	-0.08	0.03	< 0.001
Ad hoc transfers	-0.55	0.03	-0.16	0.03	< 0.001
Gifts	0.14	0.01	0.17	0.01	< 0.001
Allowances	-0.05	0.01	-0.07	0.00	0.043
Tasks	0.00	0.00	-0.01	0.00	< 0.001

Panel B: All spending

	Girls		Boys		
Category	Estimate	SE	Estimate	SE	*P* value
Total (all)	-1.90	0.03	-1.13	0.03	< 0.001
Clothing	-0.16	0.01	-0.25	0.01	< 0.001
Digital	0.23	0.01	0.90	0.01	< 0.001
F&B	-1.04	0.01	-0.85	0.01	< 0.001
Other	-0.93	0.02	-0.92	0.02	0.812

Panel C: In-person Spending

	Girls		Boys		
Category	Estimate	SE	Estimate	SE	*P* value
Total (in-person)	-2.42	0.01	-1.41	0.01	< 0.001
Clothing	-0.57	0.00	-0.04	0.00	< 0.001
Digital	-0.01	0.00	-0.02	0.00	0.002
F&B	-1.03	0.01	-0.83	0.01	< 0.001
Other	-0.81	0.01	-0.53	0.01	< 0.001

Panel D: Online Spending

	Girls		Boys		
Category	Estimate	SE	Estimate	SE	*P* value
Total (online)	0.96	0.02	0.68	0.02	< 0.001
Clothing	0.40	0.01	-0.21	0.01	< 0.001
Digital	0.24	0.01	0.92	0.01	< 0.001
F&B	-0.01	0.00	-0.03	0.00	< 0.001
Other	0.32	0.01	0.00	0.01	< 0.001

This table reports the estimated effect of the UK's initial COVID-19 lockdown on girls' and boys' income and spending. Estimates represent the change in each outcome variable between the pre-lockdown period (1 January 2019–22 March 2020) and the lockdown period (23 March–1 June 2020). In Panel B, "Total (all)" includes in-person purchases, online purchases, and ATM cash withdrawals. In Panels C and D, spending estimates are based only on debit card purchases because we can confirm the channel (in person or online) and category (clothing, digital, food, or other) for card purchases, but not for cash purchases. The model specification is detailed in Methods, 'COVID-19 analysis.' These effects are interpreted as descriptive rather than causal effects.

Extended Data Table 9a | Mediation model results: Direct, indirect, and total effects

	Direct Effect			Indirect Effect: Digital			Indirect Effect: Clothing			Indirect Effect: F&B			Indirect Effect: Other			Total Effect		
Age	Effect	SE	*P* value	Effect	SE	*P* value	Effect	SE	*P* value	Effect	SE	*P* value	Effect	SE	*P* value	Effect	SE	*P* value
6	-1.01	0.15	< 0.001	-1.54	0.06	< 0.001	0.38	0.03	< 0.001	0.49	0.10	< 0.001	0.11	0.12	0.343	-1.57	0.27	< 0.001
7	-0.95	0.11	< 0.001	-2.54	0.06	< 0.001	0.51	0.03	< 0.001	0.54	0.07	< 0.001	0.39	0.08	< 0.001	-2.04	0.19	< 0.001
8	-0.66	0.09	< 0.001	-3.89	0.07	< 0.001	0.74	0.03	< 0.001	0.73	0.06	< 0.001	0.77	0.07	< 0.001	-2.31	0.16	< 0.001
9	-0.78	0.09	< 0.001	-5.20	0.07	< 0.001	1.00	0.03	< 0.001	0.96	0.07	< 0.001	1.05	0.06	< 0.001	-2.97	0.15	< 0.001
10	-1.04	0.10	< 0.001	-6.60	0.09	< 0.001	1.62	0.03	< 0.001	1.47	0.07	< 0.001	1.41	0.05	< 0.001	-3.13	0.14	< 0.001
11	-1.31	0.13	< 0.001	-7.56	0.12	< 0.001	2.73	0.05	< 0.001	2.63	0.07	< 0.001	2.25	0.06	< 0.001	-1.25	0.15	< 0.001
12	-0.94	0.12	< 0.001	-9.09	0.10	< 0.001	4.73	0.06	< 0.001	3.73	0.08	< 0.001	3.44	0.08	< 0.001	1.87	0.17	< 0.001
13	-0.82	0.15	< 0.001	-9.92	0.12	< 0.001	6.76	0.09	< 0.001	3.92	0.09	< 0.001	3.95	0.09	< 0.001	3.89	0.22	< 0.001
14	-0.63	0.19	0.001	-9.86	0.17	< 0.001	7.69	0.10	< 0.001	3.54	0.12	< 0.001	4.11	0.11	< 0.001	4.85	0.29	< 0.001
15	-0.10	0.28	0.737	-8.89	0.20	< 0.001	7.40	0.13	< 0.001	2.61	0.16	< 0.001	3.55	0.15	< 0.001	4.57	0.39	< 0.001

This table reports results from descriptive mediation models estimated separately for each age from 6 to 15 using structural-equation modelling (SEM). Each model includes child gender (girl = 1, boy = 0), total monthly income as the outcome, and monthly spending on digital goods and services ('digital'), clothing and accessories ('clothing'), food and beverage ('F&B'), and 'other' items as parallel mediators. The direct effect is the estimated gender difference in monthly income when spending in all four categories and the control variables are held constant. Each indirect effect combines the gender difference in spending in that category with the category's association with income. The total effect is the sum of the direct effect and the four indirect effects and represents the overall gender income difference, holding the control variables constant but not controlling for spending. All effects are measured in pounds of monthly income: positive values represent differences in the direction of girls receiving more than boys, whereas negative values represent differences in the direction of boys receiving more than girls. All models control for parent age, parent age squared, parent gender, indicator variables for the number of children in the household, and year-month fixed effects. Standard errors (SE) are clustered at the household level; standard errors for indirect effects are calculated using the delta method. *P* values are from two-tailed z-tests. These effects are interpreted as descriptive associations rather than causal effects. See Extended Data Table 9b for the a-path and b-path coefficients.

Extended Data Table 9b | Mediation model results: Path coefficients

	Digital			Clothing			F&B			Other		
Age	*a*	SE	*P* value	*a*	SE	*P* value	*a*	SE	*P* value	*a*	SE	*P* value
6	-1.38	0.04	< 0.001	0.37	0.02	< 0.001	0.33	0.07	< 0.001	0.10	0.11	0.341
7	-2.24	0.04	< 0.001	0.58	0.02	< 0.001	0.39	0.05	< 0.001	0.39	0.08	< 0.001
8	-3.40	0.04	< 0.001	0.87	0.02	< 0.001	0.53	0.04	< 0.001	0.76	0.06	< 0.001
9	-4.58	0.04	< 0.001	1.22	0.02	< 0.001	0.68	0.04	< 0.001	1.05	0.05	< 0.001
10	-5.86	0.04	< 0.001	1.85	0.02	< 0.001	1.05	0.04	< 0.001	1.46	0.05	< 0.001
11	-6.84	0.04	< 0.001	2.96	0.02	< 0.001	1.85	0.04	< 0.001	2.26	0.05	< 0.001
12	-8.01	0.05	< 0.001	4.75	0.03	< 0.001	2.55	0.04	< 0.001	3.21	0.06	< 0.001
13	-8.97	0.05	< 0.001	6.62	0.05	< 0.001	2.58	0.06	< 0.001	3.76	0.07	< 0.001
14	-9.04	0.06	< 0.001	7.73	0.06	< 0.001	2.37	0.08	< 0.001	3.89	0.09	< 0.001
15	-8.56	0.07	< 0.001	8.21	0.09	< 0.001	1.73	0.10	< 0.001	3.31	0.13	< 0.001
	Digital			**Clothing**			**F&B**			**Other**		
Age	*b*	SE	*P* value	*b*	SE	*P* value	*b*	SE	*P* value	*b*	SE	*P* value
6	1.12	0.03	< 0.001	1.02	0.05	< 0.001	1.47	0.05	< 0.001	1.10	0.02	< 0.001
7	1.13	0.02	< 0.001	0.87	0.04	< 0.001	1.40	0.04	< 0.001	1.00	0.02	< 0.001
8	1.15	0.01	< 0.001	0.85	0.03	< 0.001	1.39	0.03	< 0.001	1.02	0.02	< 0.001
9	1.13	0.01	< 0.001	0.81	0.02	< 0.001	1.41	0.04	< 0.001	1.00	0.02	< 0.001
10	1.13	0.01	< 0.001	0.88	0.02	< 0.001	1.41	0.03	< 0.001	0.97	0.01	< 0.001
11	1.10	0.02	< 0.001	0.92	0.02	< 0.001	1.42	0.02	< 0.001	1.00	0.01	< 0.001
12	1.14	0.01	< 0.001	1.00	0.01	< 0.001	1.46	0.01	< 0.001	1.07	0.01	< 0.001
13	1.11	0.01	< 0.001	1.02	0.01	< 0.001	1.52	0.01	< 0.001	1.05	0.01	< 0.001
14	1.09	0.02	< 0.001	1.00	0.01	< 0.001	1.49	0.01	< 0.001	1.06	0.01	< 0.001
15	1.04	0.02	< 0.001	0.90	0.01	< 0.001	1.51	0.01	< 0.001	1.07	0.02	< 0.001

This table reports the a-path and b-path coefficients underlying the descriptive mediation results in Extended Data Table 9a. Models are estimated separately for each age from 6 to 15 using structural-equation modelling (SEM), with child gender coded as girl = 1 and boy = 0. Spending categories are digital goods and services ('digital'), clothing and accessories ('clothing'), food and beverage ('F&B'), and 'other' items. The upper panel reports a-paths: the estimated differences between girls' and boys' monthly spending in each category, measured in pounds sterling and holding the control variables constant. Positive a-paths indicate higher spending by girls; negative a-paths indicate higher spending by boys. The lower panel reports b-paths: the difference in predicted monthly income associated with an additional £1 of spending in each category, holding gender, spending in the other three categories, and the control variables constant. Within each age-specific model, each spending–income association is assumed to be the same for girls and boys. The indirect effect for each category is the product of its a-path and b-path. All models control for parent age, parent age squared, parent gender, indicator variables for the number of children in the household, and year-month fixed effects. Standard errors (SE) are clustered at the household level, and *P* values are from two-tailed z-tests. These coefficients are interpreted as descriptive associations rather than causal effects.